Elliptic-Curve Index Calculus Against Pollard Rho:

A Survey of Results, a Measured Frontier with Every Cost Priced, and Open Problems

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Abstract

Index calculus is the method that should exploit the most structure in an elliptic-curve discrete-logarithm instance, and in practice it has never beaten Pollard rho on a prime-order curve of cryptographic shape. This paper surveys where that question stands and measures it. We review the literature by field type—extension fields (Gaudry, Diem, FGHR), prime-degree binary fields and the contested first-fall-degree heuristics behind the subexponential claims (Petit–Quisquater, Semaev, and the evidence of Huang, Kosters and Yeo, Shantz–Teske and Galbraith–Gebregiyorgis), subfield curves (Galbraith–Granger–Merz–Petit) and prime fields (Petit–Kosters–Messeng, Amadori–Pintore–Sala, McGuire–Mueller)—and the generic lower bounds that frame them.

We then measure the frontier with one native harness that charges rho, baby-step giant-step, the kangaroo and several index-calculus pipelines in a single counted unit on one-target workloads with sealed, replayable records. Almost no cost in such records is complete, the rho baselines included, so we price every arm's unpriced native work alike across a band of weights and rank two methods only where the order holds across the band. On fixed ladders of eight Koblitz curves ($\log_2 r$ from 11 to 39), six NIST-shaped prime curves (20–40 bits) and five Kummer-tower curves—4,848 executions on public targets, parameters chosen on separate training targets, every session replay-audited—no index-calculus arm beats generic rho on any curve at any weight. The closest, a compact Frobenius-orbit factor base, is 3.8–15× signed-Frobenius rho with native work free; two-summand prime-field index calculus goes from 10.5× to 102× rho between 20 and 40 bits; no fitted cost exponent of a campaign index-calculus arm lies below rho's.

The measurements come from a fourteen-week engineering programme that we also document: root-finding decomposition oracles (a known mechanism; its root identity, Proposition 1, is proved in Lean, and the oracle is engineered to 383–1,535× over matrix F4 per decomposition), an $\mathbb{F}_p$ F4 brought to msolve parity on square systems, FGHR's D_4 symmetry cutting the $k = 4$ solve over $\mathbb{F}_{p^4}$ by 48×, genus-three index calculus after the cubic GHS transfer at about $q^{1.4}$ against rho's $q^{3/2}$ (the only exponent below rho's; its distance to a measured negation-map rho at $q = 2^{20}$ fell from 2.6× to 1.15× in three improvement rounds), and the negative results, including a conservation lemma for isogeny transfers and an equal-advice audit that retires the programme's online win over rho: Bernstein–Lange preprocessing rho with the same table is 53–317× faster. We close with open problems, each stated as the experiment or proof that would settle it. Nothing here concerns a deployed curve or key recovery.

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1 Introduction

For a prime-order subgroup $\langle G \rangle$ of size r on an elliptic curve $E/\mathbb{F}_q$, the best generic algorithm for the discrete-logarithm problem is Pollard rho [20, 25], with expected cost $\sqrt{\pi r/(2A)}$ group operations when A automorphisms (negation, and on a Koblitz curve the Frobenius) can be folded into the walk [11, 26]. Index calculus, the method that breaks finite-field discrete logarithms in subexponential time, was brought to elliptic curves by Semaev's summation polynomials [21] and by Gaudry's and Diem's analyses over extension fields [4, 12]. Its open question has always been *where the crossover is*: on which curves, at which sizes and under which accounting does it cost less than rho?

This paper is a survey of where that question stands, in three parts. First, the literature (§3): what is proved, what rests on heuristics such as the first-fall-degree assumption, and what the experimental record says about those heuristics. Second, a measured frontier (§11): every method family—rho variants, baby-step giant-step, the kangaroo and several index-calculus pipelines—run on the same one-target workloads

in one counted unit, on fixed curve ladders, with the native work that each method leaves unpriced priced alike on every arm across a band of weights, and a ranking reported only where it holds across that band. Third, open problems, each stated as the measurement or proof that would settle it (§12).

The measurements come from one open repository and a programme of work there between July and October 2026 (§5): about 6,900 commits and 450 dated study directories, a boundary ledger (`docs/ic/boundary_targets.json`), a native harness (`src/cryptanalysis/ecbench/`), and the engineering results of §§6–10. It is easy, in a body of work this size, to improve a headline number by two orders of magnitude and establish nothing; this paper is organised to prevent that. Every result is stated with the boundary on which it was measured (oracle stage, complete solve, cold one-target cost, online cost after retained set-up), the unit, the sample, the interval, and the committed artefact that holds the records.

The routes measured. The engineering lines, each a section below:

1. *Root-finding decomposition* (§6): replacing the Gröbner solve of the decomposition step by closed-form roots of S_3, then by meet-in-the-middle tables for more summands, then by GPU, tower and dense-index variants.
2. *Gröbner-basis engines* (§7): matrix F4, F5, an inherited F4 that re-uses its root echelon form, an index-calculus-specific F6-IC, an $\mathbb{F}_p$ F4 brought to msolve parity, and an audit that found and fixed one soundness bug.
3. *Koblitz and binary curves* (§8): Frobenius-invariant factor bases [10], their compact signed-orbit form, SAT oracles [24], and a ladder of one-target online comparisons with rho from $n = 53$ to $n = 83$.
4. *Prime and extension fields* (§9): $j = 0$ and $a = -3$ known-answer ladders, the Kummer-tower factor base of Petit–Kosters–Messeng [19], and Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$ with the D_4 symmetry of Faugère–Gaudry–Huot–Renault [7].
5. *Isogeny, transfer and structure routes* (§10): conductor gaps in the ECC2K-130 isogeny class, isogeny fibres, transfer-aware index calculus, GHS and Weil descent; mostly negative, with one lemma that explains why.
6. *Literature replication* (§9, §10): the *lit2026* programme, which turned each finding of a 2025–26 literature search into a measured change, a diagnostic, or a recorded negative.

Underneath all six sits the measurement work of §4: a single harness (*ecbench*) that charges rho, BSGS, the kangaroo and index calculus in one unit on one-target workloads, and the accounting corrections that the harness forced on earlier claims.

Contributions.

- A survey of elliptic-curve index calculus organised by field type, separating proved results from heuristic ones and recording where the first-fall-degree debate stands (§3).
- A measurement discipline: one unit (S = group-addition equivalents/$\sqrt{r}$), one-target cold and online boundaries, sealed and replayable records, and a pricing band that treats the unpriced work of rho and of index calculus alike (§4).
- A measured frontier on fixed Koblitz and prime ladders, with parameters chosen on separate training targets and fitted cost exponents per method (§11).
- The engineering record: root-finding decomposition (with its prior art), Gröbner engines, compact-orbit Koblitz bases, D_4 symmetry at $k = 4$, and the negative results (§§6–10), and the improvement ladder of 14–15 October measured against the index-calculus baseline, with the distance to rho left after each step (§11.3).
- Open problems, each with the experiment or proof that would settle it (§12).

What this paper does not claim. No asymptotic sub-rho algorithm, no change to the Gaudry–Diem picture for prime fields, no statement about ECC2K-130, P-256 or any deployed curve beyond labelled arithmetic on public parameters, no key recovery. Every instance is a public synthetic or standard-parameter known-answer instance. Every change is classed, in the repository's vocabulary, as an ADVANCE (an exponent or a crossover moves under complete accounting), ENGINEERING (a constant moves), RELABELLING, or ACCOUNTING; with one modelled exception (§9.4) every improvement in this paper is ENGINEERING or ACCOUNTING.

How to read the numbers. "$X\times$" always means baseline cost divided by candidate cost on the stated boundary. Stage ratios (one decomposition, one Gröbner call) are never presented as end-to-end speed-ups; wall-time ratios from hosts that did not earn an isolation level are labelled *descriptive*; a cost with unpriced work is labelled a *lower bound*. Appendix A lists the artefact behind each table.

2 Background

Rho and its floor. With A efficiently computable automorphisms acting on $\langle G\rangle$, a rho walk on orbit representatives has expected cost $\sqrt{\pi r/(2A)}$ group additions. $A = 2$ (negation) on every curve; $A = 2n$ on a Koblitz curve $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, where the Frobenius $\pi(x, y) = (x^2, y^2)$ acts on $\langle G\rangle$ as a scalar λ [11, 26]. ECC2K-130 is the case $a = 0$, $n = 131$ of this family; its record GPU rho attempt is a separate paper in this repository (`paper/ecc2k130/`).

Index calculus on elliptic curves. Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$. A *relation* is a decomposition $aG + bQ = R = P_1 + \cdots + P_m$ with $P_i \in \mathcal{F}$; once about $|\mathcal{F}|$ relations are found, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. The pipeline has five stages (factor base, decomposition oracle, relation yield, rank/linear algebra, verification) and a sixth, the comparison with rho on the same target. The *decomposition oracle* decides whether R splits over $\mathcal{F}$. Semaev [21] reduced it to the roots of the summation polynomial $S_{m+1}(x_1, \ldots, x_m, x_R)$, which vanishes exactly when $\pm P_1 \pm \cdots \pm P_m = R$ for points with those abscissae. Over $\mathbb{F}_{2^n}$ or $\mathbb{F}_{p^k}$ the factor base is chosen as the points whose abscissa lies in a subspace or subfield, and Weil restriction turns $S_{m+1} = 0$ into a polynomial system over the base field; the textbook oracle solves that system with a Gröbner-basis engine (F4/F5, [5, 6]) or a SAT solver [24]. Gaudry [12] and Diem [4] give the asymptotics over extension fields; for prime fields no factor base is known that makes the method competitive, and summation-polynomial systems over $\mathbb{F}_p$ are the subject of the Kummer-tower and rational-map bases of Petit–Kosters–Messeng [19].

Frobenius-invariant bases. Galbraith, Granger, Merz and Petit [10] observe that a factor base with $\pi(\mathcal{F}) = \mathcal{F}$ splits into Frobenius orbits of size dividing n, with $\log \pi^k(P) = \lambda^k \log P$. The unknowns collapse by $\approx n$, the matrix by $\approx n^2$, and ordered decompositions save $m!$—more than rho's $\sqrt{2n}$, but from a base too far behind to overtake it at deployed sizes. Invariant bases come from linearised polynomials attached to the factors of $x^n - 1$ over $\mathbb{F}_2$; the subspace dimension is $\ell = \mathrm{ord}_n(2)$, so the base size is fixed by n (Table 1 of `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`).

Symmetries of the system. The decomposition system is invariant under the symmetric group S_m; rational 2-torsion adds sign changes, and Faugère, Gaudry, Huot and Renault [7] show that in the coordinate Y with $Y(P + T) = -Y(P)$ the invariance group becomes the Weyl group D_m, with a much smaller quotient. We measure this at $m = 4$ in §9.4.

3 The theory, by field type

3.1 What a decomposition costs

Every index-calculus algorithm for elliptic curves in the literature spends its time in three places: finding about $|\mathcal{F}|$ relations (each a decomposition of a random combination $aG + bQ$ over $\mathcal{F}$), solving the sparse linear system over $\mathbb{Z}/r$, and, in some variants, a descent of the target. The second and third are well understood. The first is the crux: a uniformly random point decomposes over a base of size N into m summands with probability about $N^m/(m!\, r)$, so relations need about $m!\, r/N^{m-1}$ trials, and each trial costs whatever the decomposition oracle costs. Every result below is a statement about one of three things: how large N can be while membership stays algebraically testable, how cheap the oracle is, and how the two trade against the linear algebra.

3.2 Extension fields $\mathbb{F}_{q^k}$ with $k > 1$ fixed or small

Gaudry [12] takes $\mathcal{F}$ to be the points with abscissa in the base field $\mathbb{F}_q$ and decomposes into $m = k$ summands by Weil restriction of S_{k+1}, a system of k equations in k unknowns over $\mathbb{F}_q$. For fixed k this gives $\tilde{O}(q^{2-2/k})$, asymptotically below rho's $q^{k/2}$ for $k \geq 3$; the constant hidden in the Gröbner solve grows very quickly with k, which is what makes the crossover large in practice (to be quantified from the paper). Diem [4] extends the analysis to families where k grows, proving subexponential complexity over sequences of fields with suitably balanced q and k. Joux and Vitse's work on decomposition and cover attacks over small extension fields is to be added here once read. FGHR [7] exploit the action of S_m and of the 2-torsion to replace S_{m+1} by its invariant form, cutting the Gröbner work by large factors; our measurement at $k = 4$ (§9.4) is a direct instance.

3.3 Binary fields $\mathbb{F}_{2^n}$ with n prime

Prime n leaves no subfield, so the factor base is taken to be the points whose abscissa lies in an $\mathbb{F}_2$-vector subspace V of dimension ℓ. Weil restriction then gives a quadratic-to-low-degree Boolean system whose difficulty is governed by its degree of regularity. The whole debate about binary curves is a debate about that degree.

The subexponential claims. Petit and Quisquater [18] conjectured, from experiments and heuristic evidence, that these systems have degrees of regularity only slightly above the degrees of their equations (the *first fall degree assumption*), and deduced a heuristic $2^{c\,n^{2/3}\log n}$, $c < 2$, for a variant of Diem's algorithm, beating generic methods only for $n \gtrsim 2000$. Semaev [22] split summation polynomials into cubic Boolean systems, argued for regularity degree at most 4 under the same kind of assumption, and obtained a heuristic $2^{c\sqrt{n \ln n}}$ with $c \approx 1.69$; its first version concluded that several FIPS binary curves fall faster than Pollard rho, a conclusion its revision says does not hold under the weaker assumption that the regularity degree is $o(\sqrt{n/\ln n})$.

The evidence against. Shantz and Teske [23] tested the Petit–Quisquater heuristics experimentally. Huang, Kosters and Yeo [15] introduced the *last fall degree*, independent of the monomial order, gave computational evidence against the first fall degree assumption in characteristic 2, and constructed summation-polynomial Weil descent systems where it is unlikely to hold. Galbraith and Gebregiyorgis [9] improved the variable choice (binary Edwards curves), found symmetry-breaking factor bases and SAT solvers inconclusive, and concluded that rho remains much faster for prime-degree binary fields of reasonable size. The Galbraith–Gaudry survey [8] reaches the same verdict for characteristic-two curves over prime-degree fields.

Where it stands. No proof or refutation of a subexponential algorithm for prime-degree binary curves exists. The open quantity is the solving degree of Weil-descended S_{m+1} as n and m grow; recent work supplies diagnostics (saturation-exponent bounds, multi-graded regularity, tests for where systems stop looking generic) but no bound for these systems. Our own registered refutation-degree law at $m = 3$ (§7) is one data point and ended inconclusive.

3.4 Subfield (Koblitz) curves

When the curve is defined over $\mathbb{F}_2$ the Frobenius acts on the group, and Galbraith, Granger, Merz and Petit [10] show that a Frobenius-invariant factor base saves a factor $\approx n$ in relations and $\approx n^2$ in linear algebra—more than rho's $\sqrt{2n}$—while concluding that Koblitz-curve security is not threatened at deployed sizes. Our Koblitz work (§8) builds on this.

3.5 Prime fields

Over $\mathbb{F}_p$ there is no subfield and no subspace, and no factor base is known for which membership is both dense enough and algebraically testable. Semaev's original small-abscissa base leads to systems of very high degree. Petit, Kosters and Messeng [19] construct bases from rational maps (including the Kummer tower μ_{2^t}) that give the summation system extra structure; Amadori, Pintore and Sala [1] build the base from the instance and need only one relation search, improving on both; McGuire and Mueller [17] avoid Gröbner bases altogether by fast evaluation of S_m through iterated resultants, and in a second variant by summing $m-1$ factor-base points and testing membership of the result—the same mechanism as the two-summand root-finding oracle of §6. All of these remain exponential and slower than rho; a 2024 analysis again concludes that no subexponential prime-field method is known.

3.6 Lower bounds

Generic-group lower bounds bound everything that does not use curve structure. With preprocessing, Corrigan-Gibbs and Kogan [3] show $ST^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T; this is the bound against which the retained-set-up (online) results of §8 must be read. A 2026 structured-generic-group bound shows that an algorithm exploiting structure on only a δ fraction of elements needs $\Omega(\min(\sqrt{q}, 1/\delta))$ time—for index calculus, the decomposable set must be dense *and* cheaply testable.

4 Measuring index calculus against rho

The programme's first lesson was that index-calculus numbers from different studies did not compose: one study timed a decomposition, another a relation, a third a whole multi-target batch with shared tables, each on a different host and in a different unit. Four mechanisms, built in this order, fixed that.

4.1 The boundary ledger (10 September)

`docs/ic/BOUNDARY_TARGETS.md` and its machine twin `docs/ic/boundary_targets.json` give each regime (Koblitz, binary, prime) and each pipeline stage (`factor_base`, `decomposition`, `relation_yield`, `rank`, `end_to_end_dlp`, `vs_rho`) a current record, a next target and acceptance gates. A row promotes only when its evidence JSON carries every field the measurement schema requires for that stage—for example, a `decomposition` claim that beats an algebraic frontier must report the first-fall degree, and a `vs_rho` claim needs one target, the identical public point for both methods, verified scalars, five exclusive IC online phases, the automorphism discount, matched resources and a host-isolation receipt. Missing fields fail closed. The primary `vs_rho` question is the online cost of *one* previously unseen target after reusable IC preparation; multi-target and amortised results are secondary and must be declared as such.

4.2 ecbench (from 2 October)

ecbench (`src/cryptanalysis/ecbench/`, `src/bin/ecbench.rs`, documented in `docs/ecbench/README.md`) runs every ECDLP method behind one interface, on the same single-target workloads, in one unit, into sealed records.

Unit.

$$S \;=\; \frac{\text{total group-addition equivalents (GAE)}}{\sqrt{r}}, \qquad S_{\text{floor}} = \sqrt{\pi/(2A)}.$$

An addition or doubling is one GAE; a scalar multiplication is charged as the additions it performs; native work (lookups, row operations, square roots, Artin–Schreier solves, word XORs) is priced at a pinned conversion where the repository has one and is otherwise recorded as *uncharged*, which marks the record's S as a *lower bound*. Plain rho has $S \approx 1.25$.

What a measurement is. One workload is one curve, subgroup, generator and target; nothing survives between executions, so every table a method builds is inside its cost (*cold*). Beside the cold S every record carries the one-target *online window*, split for index calculus into the five exclusive phases `target_query`, `target_PDP`, `target_relation_check`, `target_descent`, `target_recovery_check`. Targets are either planted or *public*—hashed to the subgroup so that nobody knows the logarithm—and the runner verifies $[k]G = Q$ in its own process; `ecbench verify` checks it again from the files. Failed, exhausted and timed-out runs are rows, never dropped.

Identity, sealing and replay. Every object (curve, factor base, method parameters, spec, session, host environment class) is named by a hash of what it is (`ECBS1h...` for sessions, `ECBND1h...` for bounds). Each execution appends one sealed record; `ecbench verify` replays measured records and emits an audit receipt whose SHA-256 is the replay certificate. Operation counts are deterministic, so a replay must match to the last digit; a mismatch is preserved as a failed audit (§4.5).

Isolation. A run earns L0 (recorded under a host capsule), L1 (pinned, affinity read back, no foreign SMT sibling, CPU/wall ≤ 1.05), L2 (quiet: zero steal, foreign busy time $\leq 1\%$, run-queue delay $\leq 0.5\%$, ≤ 50 preemptions/s) or L3 (isolated cores, fixed governor, bare metal). Wall time is admitted only at the spec's required level (default L2) with at least five admitted pairs and an A/A interval; otherwise it is printed as *descriptive*. Operation counts need no isolation. macOS hosts earn L0; a GitHub-hosted runner earned L1 on 72/72 runs and matched a macOS session's operation counts to the last digit. *isolab* prepares lab hosts for L3 and cross-host replay.

Statistics. The comparable statistic is the mean S (the floor is an expectation). Comparisons are paired by (workload, round) with shared seeds, reported as a ratio of totals $\sum S_B / \sum S_A$, never a mean of ratios, with a two-stage cluster bootstrap that resamples workloads, then pairs. An operations A/A is exactly 1 by construction and checks determinism. A claim about scaling fits exponents over at least four sizes.

Bounds and the frontier. `ecbench` also records sealed *bounds*—constant and exponent with intervals, per phase—and a challenge protocol whose verdict says whether a candidate advances, trades, matches or regresses an incumbent and at which level (exponent, constant, primitive). The incumbent for cold one-target Koblitz operations is `rho.signed_frobenius_strong` (bound `ECBND1h05a2ea918f3c`): $1.089\times$ the floor $[0.927, 1.261]$, fitted exponent $\alpha = 0.451$ $[0.399, 0.512]$, 144 runs at six sizes (`research/ecbench_frontier_20261013/`).

Definition 1 (comparability)**.** A performance point is keyed by the tuple

$$D = \big(\text{curve and subgroup},\ \text{target law},\ \text{cold} \mid \text{online} \mid \text{stage},\ \text{unit and unpriced work},\ \text{set-up amortisation } q,\ \text{memory envelope},\ \text{correctness/replay/isolation class}\big).$$

Two points are ratio-comparable only when every component of D except the method matches; a method is on the frontier of a domain when no admitted point is no worse on every axis and strictly better on one.

This rules out the four collapses we saw most often: online wall as cold cost; word XORs, probes, field operations and GAE as interchangeable; a lower bound as a complete cost; an L0 wall ratio as an admitted speed-up.

4.3 The pricing band

Almost no cost in a record is complete, and the rho baselines are no exception. Rho, BSGS and the kangaroo leave canonicalisations, partition hashes and table inserts and lookups unpriced. An index-calculus arm is complete only on curves that have a pinned timing calibration (`docs/ic/calibration.json`), which converts lookups, row operations, Artin–Schreier solves, Frobenius maps and solver word XORs to GAE at measured ratios of roughly 10^{-3} to 10^{-1}. A "complete" index calculus set against a lower-bound rho is therefore lopsided, and several earlier comparisons in this repository were of that kind.

Since 15 October every cross-method comparison uses a *pricing band* (repository rule, `AGENTS.md` §5a; native implementation `ecbench band`). For a record with charged cost G, calibration-priced native cost C over n_C operations, and U operations behind its unpriced counters,

$$S(w) = \frac{G - C + w\,(U + n_C)}{\sqrt{r}}, \qquad w \in \{0,\ 0.1,\ 1\} \text{ GAE per native operation},$$

applied alike to every arm, references included. Two arms are ranked only if their order is the same at every weight; otherwise the pair is reported as unresolved. An arm that names an unpriced counter without counting it is a lower bound and cannot be ranked. The band brackets the pinned calibrations from below and above, so a ranking that survives it does not depend on the host the calibration came from.

4.4 Claim hygiene and change classes

Every study directory states a protocol before it runs (frozen inputs are committed first), keeps failures, and closes with a decision in one of the four classes of §1. A controlled-experiment rule requires a negative control from the right null population, a positive control that proves the detector can see the effect, and the same instrument on every arm before any comparative statement. Blockers are recorded as named problems (`problems/`) with exit criteria rather than silently dropped. Measurement code is native (Rust); Python only lays out recorded values.

4.5 Corrections the accounting forced

Building the harness and the ledger exposed errors in earlier claims, including our own. We list them because they change how earlier numbers must be read.

1. *Online Koblitz ratios relabelled* (7–8 October, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`). Same-point online ratios of 124.85× ($n = 61$), 2,864.1× ($n = 71$), 1,189.7× ($n = 73$) and 8.2× ($n = 83$) had verified answers but no host-isolation receipt; they are *exploratory*. Only the later 64-target pinned panel at $n = 61$ is admitted (§8).
2. *ECC2K-130 extrapolation constants corrected* (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`): the structural model predicts $C = 0.25$ in probes/relation $\approx C\,r/(n^2K^2)$; the fitted C is 0.27 at $n = 61$ but 1.2–1.5 at $n = 71, 73$, an unexplained gap now stated rather than absorbed.
3. *"F4" that was Buchberger* (`research/ecbench_rootfinding_assessment_20261012/REPORT.md`): the NIST benchmark of the root-finding manuscript printed its baseline as F4 but called `groebner_f4::buchberger`; its 3,021–5,802× are ratios to Buchberger. The manuscript's claim that no implementation can use fewer than linear field operations per ideal is too broad once a reusable pair table is allowed.
4. *Unpriced word XORs* (`research/ecbench_rootfind_faster_20261014/`). Inherited F4's Boolean word XORs were counted but not charged, so its $S = 16.7$ at Koblitz $n = 13$ was a lower bound that looked competitive with rho (14.4). Pricing them gives $S = 421.4$.
5. *Cross-host counter drift* (`research/ecbench_frontier_20261013/audits/`): a replay-all audit of the root-finder session on x86_64 matched 120/168 records; all 48 inherited-F4 rows differed in solver word-XOR counts (e.g. 301,961 versus 300,531), with every scalar, trial count and charged GAE unchanged. The failed receipt is kept; the F4 stage comparison is historical until the drift is explained.[1]
6. *A soundness bug* (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` §2.1): a degree-bounded `f4_fp::solve` could certify solutions of an inconsistent system (over $\mathbb{F}_5$, $\{x^3 + 1, xy - 1, x + y\}$ returned two "solutions"). Fixed by evaluating every candidate on the input.
7. *F6-IC charged cost* (§7): a stage win in word XORs became a 64.6–1,353× loss in charged S once its batched geometry was counted—recorded as an accounting result, not an advance.
8. *Lopsided pricing* (§4.3): comparisons of a calibration-priced index calculus with a lower-bound rho. On the frontier campaign's train panel at Koblitz $n = 13$ the compact-orbit arm, earlier quoted at 1.08× rho as a lower bound, is 0.93×, 1.53× and 5.7× the strong rho at $w = 0$, 0.1 and 1 (columns 2, four train targets, `research/ic_frontier_campaign_20261015/select/SELECTION.md`): below rho with native work free, above it otherwise, so the pair is unresolved inside the band rather than a win for either side.

5 Chronology

Figure 1 shows commit activity on `main` by ISO week. Theme lines are case-insensitive keyword matches on commit subjects (a commit may match several themes or none; the regular expressions are in the paper's `README.md`), so they indicate where effort went, not how much of it was index calculus. The grey bars include unrelated work in the same repository (the ECC2K-130 GPU rho campaign, an isogeny KEM, post-quantum benchmarks), which dominates week 37.

Prelude: July and August. The prime-field *ecdlp autolab* ran numbered threads on curve structure—universal order-2 Frobenius forms, the CM-73 family $\{19, 37, 79, 109\}$, sextic-twist pairs that are Howe-glueable (5 of 15)—and its draft "An Empirical Resistance Map for Prime-Field ECDLP" (`research/ecdlp_autolab/PAPER_DRAFT.md`). Those threads found structure but no solver advantage, which is what turned the programme towards index calculus, where structure is supposed to pay.

Phase 1, 6–14 September: Koblitz index calculus from scratch. The ECC2K-130 rho client and an audit of a public index-calculus benchmark's SAT/UNSAT labels (6 Sept) were followed by Frobenius-invariant factor bases with a matrix-F4 Semaev oracle, an XOR-native SAT oracle after Trimoska et al. (7 Sept), and a $\sim 50\times$ faster Semaev solve the same day. On 8 September a scaling experiment showed

[1] Erratum 2026-10-20 (validation sweep A, verdict: contradicted). The 120/168 figure is the 2026-10-13 receipt. Replaying the same session on current main gives 72/168 identical, with 96 mismatches in all four IC arms (ic-f4-m2, ic-f4-m3, ic-rootfind-m2, ic-rootfind-m3); this is an open regression (sweep D, PR #2290), and the root-finder rows are not claimed to pass.

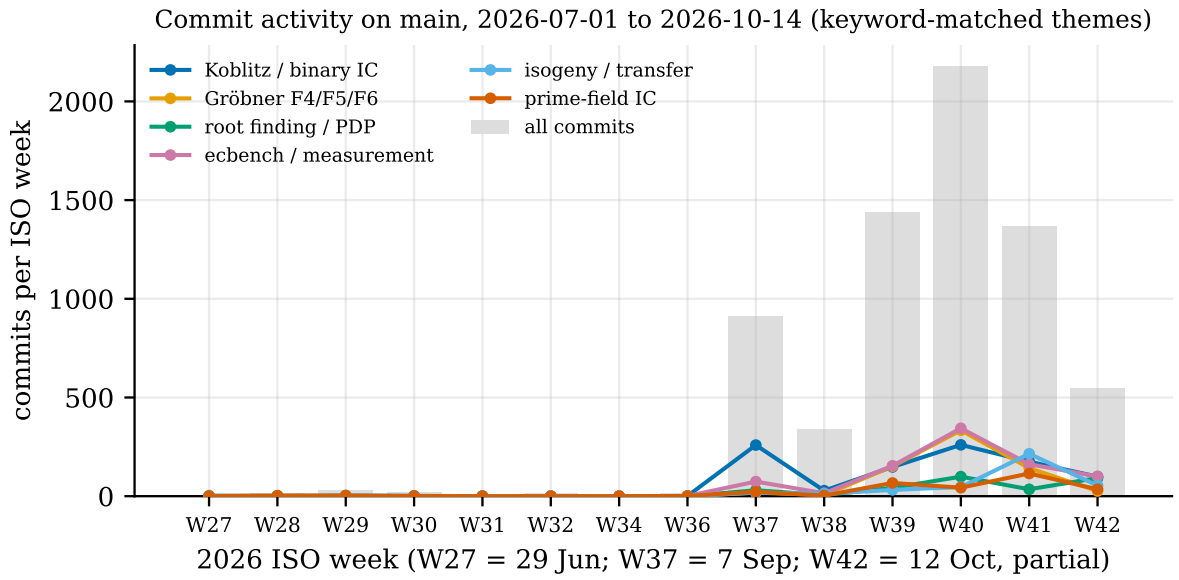


Figure 1: Commits per ISO week on `main`, 1 July–14 October 2026 (6,882 commits; week 42 partial). Lines: commits whose subject matches each theme's keywords. Source: `git log` at the paper's base commit; values in `figures/data.json`.

the SAT route does not scale and that the refutation frontier belongs to F4; pairs-and-solve replaced triple enumeration for S_4. The boundary ledger was codified on 10 September, together with a meet-in-the-middle oracle, incremental modular linear algebra and a CADO-style split into factor-base logs then descent. A burst of more than 100 numbered pipeline stages (12–13 Sept: PCLMUL field products, fused pair collection, scalar-blind targets, rank-aware scans) took the multi-target $n = 53$ pipeline to a 1,024-target process-wall crossover against batched rho. On 13–14 September six pre-registered ECC2K-130 decomposition experiments ran ("exists, admissible, not findable"; no falsifier fired), and the repository began to require frozen benchmarks and full-pipeline evidence for any ECDLP speed claim.

Phase 2, 15 September–1 October: engines and the compact orbit. Inherited F4 (22 Sept) became the default algebraic oracle; a fast binary-field hot path, Itoh–Tsujii inversion and a parallel radix-sorted pair-table build landed (23–24 Sept); a long Boolean F4/F5 kernel series (AVX2/AVX-512 unpackers, compact columns, pivot orders; 26 Sept–3 Oct) ran in parallel. The compact signed-Frobenius orbit producer beat batched rho on multi-target $n = 61$ batches (25–26 Sept), and from 2 October carried the one-target online ladder to $n = 71$ and (6 Oct) $n = 73$ on 128-bit words, after $n = 63$ was closed as non-viable. A fixed-policy cold comparison on orbit-disjoint public points (1 Oct) was a measured no-go.

Phase 3, 2–10 October: one harness, and the audit it forced. *ecbench* landed on 2 October. F6-IC and a 23-study $n = 83$ campaign (4–6 Oct), the registered $m = 3$ refutation-degree law (5–11 Oct), the ECC2K-130 feasibility extrapolation (5 Oct), the accounting correction plan, conductor-gap and novel-direction notes (7 Oct), and the transfer-aware, isogeny-fibre and Gröbner-engine audits (10 Oct) followed.

Phase 4, 11–14 October: literature, root finding, consolidation. The lit2026 programme (11 Oct) turned a literature search into twelve workstreams. A supplied manuscript on root-finding decomposition was frozen, audited and re-measured in ecbench (12 Oct), and the identity it rests on was generalised across PDP oracles, a CUDA kernel, Kummer towers, Koblitz engines and dense root indices (12–14 Oct). The admitted $n = 61$ online panel ran on two hosts (12 Oct). Gaudry's $k = 4$ solve was made generic and given the D_4 symmetry; the frontier synthesis and its replay audit closed the period (13 Oct); the last root-finding loops priced F4's word XORs and ran out of levers against rho (14 Oct).

Phase 5, 14–15 October: measured against the baseline. Two improvement rounds and a next-steps campaign paired each change against the index-calculus incumbent under frozen protocols (§11.3):

genus-3 index calculus after the cubic GHS transfer (three rounds), the $k = 4$ ladder and its four-unknown solve, production free relations and oracle row harvesting on Koblitz curves, and a CADO-NFS-derived filter for the linear algebra. An equal-advice audit retired the online win (14 Oct). The owner then fixed the success criterion as iterative improvement over that baseline, with the gap to rho as distance remaining (15 Oct), and the priced frontier campaign of §11.1 ran the same day.

6 Root-finding decomposition

6.1 Origin: a suggestion in an audited manuscript

The textbook decomposition oracle builds Semaev's polynomial system and solves it with a Gröbner-basis engine. On 12 October a 13-page manuscript, *Point Decomposition by Root Finding*, was supplied for assessment. We froze it (`research/ecbench_rootfinding_assessment_20261012/source-paper.pdf`, SHA-256 `743d3d3e...7c01`) and audited it against its source branch before measuring anything. Its central suggestion is simple: for two summands, fixing one factor-base point turns S_3 into a quadratic in the remaining variable whose roots are already known.

Proposition 1 (roots of S_3)**.** *Let $E : y^2 = x^3 + ax + b$ over a field K of characteristic $\neq 2, 3$, and let $R, F \in E(K)$ be affine with $x_R \neq x_F$. Then*

$$S_3(x_R, x_F, X) = (x_R - x_F)^2 \, (X - x_+)(X - x_-), \qquad x_\pm = x(R \pm F),$$

so $S_3(x_R, x_F, X) = 0$ iff $X \in \{x(R + F), x(R - F)\}$.

Proof sketch. With $\lambda_\pm = (\pm y_F - y_R)/(x_F - x_R)$ the chord slopes, $x_\pm = \lambda_\pm^2 - x_R - x_F$. Expanding $(X - x_+)(X - x_-)$ and eliminating y_R^2, y_F^2 with the curve equation gives exactly Semaev's $S_3/(x_R - x_F)^2$. The identity is machine-checked over an arbitrary field in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`, theorems `SemaevS3.S3_factor` and `SemaevS3.S3_root_iff`; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`). □

Hence the two-summand system has a parametrised variety: $R = \pm F_i \pm F_j$ iff $x(R \mp F_i) \in x(\mathcal{F})$. No polynomial system is solved; the oracle is one subtraction and one table probe per base point.

Prior art. Neither the identity nor the oracle is new. The identity is a direct consequence of how S_3 is defined: it is the polynomial whose roots are the abscissae of $P_1 \pm P_2$. The oracle mechanism—sum $m - 1$ factor-base points in the group and test whether the result lies in the base—is the second algorithm of McGuire and Mueller [17], who also evaluate S_m by iterated resultants instead of solving it, building on the instance-derived bases of Amadori, Pintore and Sala [1]. What this section adds is engineering and measurement: a machine-checked proof, a fail-closed verification protocol, binary pair-sum tables for up to four summands, GPU, Kummer-tower and dense-index variants, and complete-cost measurements in one harness against rho.

What the audit kept and corrected. The identity is exact under the stated generic affine condition, and a returned witness can be re-added in the group for soundness. Completeness of the exhaustive enumeration holds only with an explicit sign and factor-base convention (the binary table stores full point keys over a base containing both signs; the manuscript's schematic x-only table needed that convention stated). The manuscript's NIST baseline was Buchberger, not F4 (§4.5), and its statement that no implementation can use fewer than linear field operations per ideal fails for repeated queries against a reusable pair table. Larger-summand enumeration has a growing exponent in $|\mathcal{F}|$, and the manuscript did not benchmark it end to end. All of this is recorded in `research/ecbench_rootfinding_assessment_20261012/REPORT.md`.

6.2 The algorithms

Prime fields (Algorithm 1, `src/cryptanalysis/prime_fast.rs`): $\Theta(N)$ field operations and $2N$ lookups per target. *Binary fields* (`src/cryptanalysis/koblitz_index_calculus.rs`, `DecompositionStrategy::PairTable`, made the default): a precomputed table of pair sums $F_i + F_j$ turns an m-decomposition into 1, N or N^2 lookups for $m = 2, 3, 4$—the meet-in-the-middle half of a birthday search, exact and complete like exhaustive search, at the price of $\Theta(N^2)$ set-up and memory.

Algorithm 1 Prime two-summand oracle by direct subtraction

Require: target R, base $\mathcal{F} = \{F_1, \ldots, F_N\}$, hash set H of abscissae (or full points) of $\mathcal{F}$

1: $d_i \leftarrow x_R - x_{F_i}$ for all i; invert all d_i with one field inversion and $3(N-1)$ multiplications (Montgomery's trick)
2: **for** $i = 1..N$ **do**
3: **for** $\varepsilon \in \{+1, -1\}$ **do**
4: $\lambda \leftarrow (y_R + \varepsilon y_{F_i})\, d_i^{-1}$; $x_3 \leftarrow \lambda^2 - x_R - x_{F_i}$ ▷ $x(R - \varepsilon F_i)$
5: **if** $x_3 \in H$ **then** recover the sign of F_j from y_3; emit $R = \varepsilon F_i + \tau F_j$
6: **end if**
7: **end for**
8: **end for**

Table 1: Root-finding generalisations, 12–14 October. Every row is ENGINEERING: no slope fell below its generic floor.

Step	What it adds; artefact
Prime subtraction, binary pair table	Prop. 1; batched inversion; $m \leq 4$ tables. `research/semaev_root_vs_groebner_20261012/`
NIST P-192…P-521	standard parameters, 6- and 64-point bases; dedicated P-256 field 11–12× over generic big integers. `research/semaev_rootfind_nist_20261012/`
End to end vs msolve	complete DLPs with only the oracle swapped. `research/semaev_rootfind_e2e_20261012/`
PDP oracle family	`rf-mitm`, `mitm3`, `mitm4` (2-sum table); `rf-pairs`, `pairs3` (pairs then solve by the group law); `subtract2`; binary rungs $\ell = 8..14$, prime Kummer towers $m = 2, 3, 4$. `research/pdp_root_finder_20261012/`
CUDA kernel	one thread per (target, slice); per-thread Montgomery inversion; SoA base; shared-memory hash for $\lvert\mathcal{F}\rvert \leq 2048$; 64-bit and P-256 fields. Validated in host emulation only (no GPU available). `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`
ecbench engines	`pdp3-koblitz:m=2|3,engine=rootfind`, `rootfind-subtract`, later `rootfind-window` and `rootfind-negfold`. `research/ecbench_rootfind_engine_20261012/` and successors
Kummer-tower base	PKM's μ_{2^t} base with the root-finding oracles, 20–28-bit primes. `research/pkm_tower_ecbench_20261013/`
Dense root index, $n = 83$	packed five-byte S_3 root table, prefix widths K1271–K1798, 8 GiB of slots. `research/n83_dense_root_20261013/`
Prime tuning	pair-table size, arity, batched table inversions, synchronised lanes. `research/ecbench_rootfinding_\{tuning,arity,batchtable,lanes\}_2026101*/`

Both are verified fail closed: every decomposition is re-added in the group, S_3 is checked to vanish, Gröbner-variety membership and agreement on refutation are checked against F4, and planted positive controls must be found.

6.3 Generalising the observation

Over three days the identity was carried through every layer of the pipeline. Table 1 lists the steps; each is its own study with a protocol registered before it ran.

6.4 Results

Oracle stage. Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $\lvert\mathcal{F}\rvert = 55$, 254 targets) and 1,535× ($\mathbb{F}_{2^{13}}$, $\lvert\mathcal{F}\rvert = 4{,}005$, 256 targets) faster than matrix F4; 3.5–$4.7 \cdot 10^4$× faster than `groebner_f4` on 16–24-bit prime curves ($\lvert\mathcal{F}\rvert = 6$, 40 targets each); and 3,021–5,802× faster than Buchberger on the decomposition ideal of the five NIST curves (Figure 2). Every binary verdict agreed with F4; every prime solution was in F4's variety and every prime refutation was F4's unit ideal. In the PDP study, `rf-mitm` refutes a binary target in 3.1 μs at $\ell = 8$ and 482 μs at $\ell = 14$, against the PS-0 baseline's 9.3 ms and 55.5 s (3,007–115,160×), costs $5\lvert\mathcal{F}\rvert - 3$ multiplications and $2\lvert\mathcal{F}\rvert$ lookups a target, and sits on the pair-table floor at 1.000 ± 0.027 bits per rung; at prime $m = 3$, $N = 18$ it uses 197 multiplications where F4 uses $8.99 \cdot 10^{13}$ multiply-adds; 707/707 brute-force agreements (`research/pdp_root_finder_20261012/`).

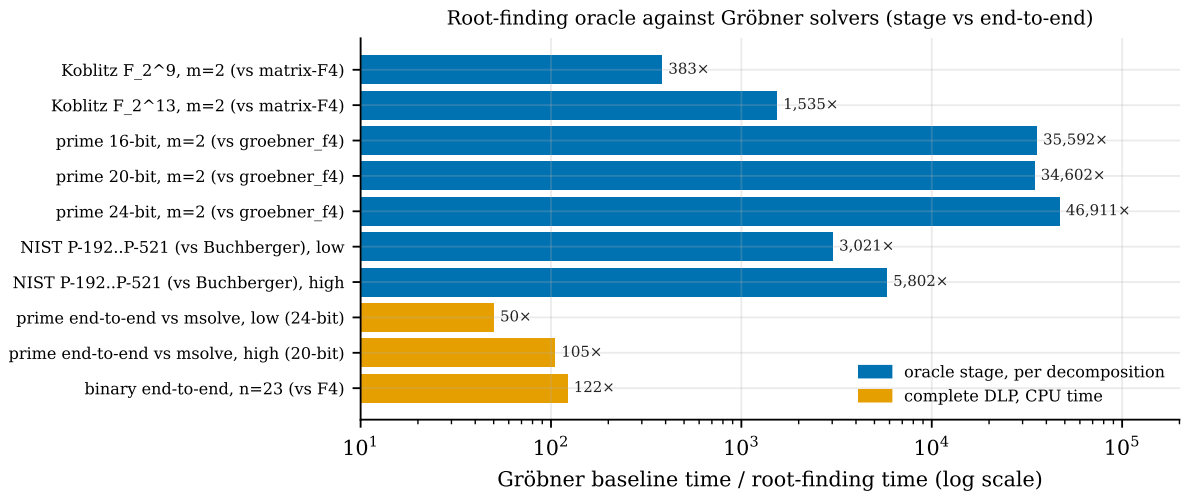


Figure 2: Root finding against Gröbner solvers on identical targets. Blue: one decomposition (oracle stage only). Orange: complete discrete-log computations with the oracle swapped and everything else fixed (prime: CPU time against msolve, 20/24-bit totals composed from sampled per-query cost; binary: one cold solve including table build). Sources: `RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`, `papers/rootfinding-decomposition/README.md`.

Complete solves. End to end, the gain is two orders of magnitude smaller and not one-sided: 50–105× against msolve on 16–24-bit prime curves; 122× on the one binary instance ($n = 23$) large enough to amortise the pair table; and F4 *faster* at $n = 5, 9, 13$, where a solve needs too few decompositions to pay for the table. Inside ecbench, with every group operation charged, the root-finding IC beats inherited F4 in solve wall by about 350× at $n = 13$ (admitted) and about 10^4× at $n = 23$ (descriptive), yet stays 2.9–25,116× the strong rho in S on binary curves and 29–141× rho on prime curves, with the gap widening (`research/pdp_root_finder_ecbench_20261013/`).

The Koblitz loops. Figure 3 follows the best complete-count root-finding arm at Koblitz E_0, $n = 13$ through four registered improvement loops (8 public targets × 3 rounds each, L0 operation counts). The first engine had $S = 38.82$ ($m = 3$) and 28.38 ($m = 2$). A windowed third-summand scan cut $m = 3$ by a paired 0.828 $[0.783, 0.863]$; a fold over negation (`rootfind-negfold`) took $m = 2$ to 23.22 and then 22.97. Two further cycles of denser and sparser bases, windowed negfold and plain root finding were dead ends (`research/ecbench_rootfind_vs_rho_20261014/`). With F4's word XORs priced, the paired ratio negfold/F4 is 0.0545 $[0.047, 0.064]$, a complete-against-complete comparison; against strong signed-Frobenius rho ($S = 14.82$) the best complete arm is still 1.55×. A compact-orbit base with a counted Frobenius pair table reaches $S = 15.55$, below the then-unpriced F4 (16.75), but it is itself a lower bound (factor-base construction natives unpriced), and it is 1.08× rho. Amortised over 64 targets that arm reaches $S \approx 11.2$, below rho; by the ledger's rules that is a secondary, multi-target result, not a cold claim.

Prime tuning in ecbench. On a 20-bit prime curve, the pair table's counted S is 0.018861 $[0.018547, 0.019242]$ of direct subtraction (24 pairs; prime native work unpriced, so a bounded ratio), and 0.057047 $[0.041159, 0.092069]$ on a 17-bit Koblitz curve, fully priced, where the table set-up is about 68% of the arm's cold cost. Doubling the table from 32 to 64 abscissae gives 0.830 $[0.697, 0.962]$; three summands instead of two costs 1.431× $[1.225, 1.716]$; batching the table inversions leaves counted GAE unchanged (1.000; 4,096 versus 64 inversions, set-up wall 243 versus 128 μs, descriptive); and a train-set lane selection did not resolve a hold-out winner (0.979 $[0.919, 1.038]$; 1,088/1,088 verified executions) (`benchmarks/README.md`, rows of 12–14 Oct).

Dense root index at $n = 83$. Widening the prefix base from K1500 to K1798 cut mean probes per relation from 300,021.9 to 169,069.8 (paired difference 130,952, 95% [52,058, 209,991]) and gave a one-point online ratio of 18.528 over same-point rho (IC 63.3 ms, rho 1,173.3 ms), with 339.0 s of reusable preparation and an 8 GiB table outside the window. The host was shared, so the result is exploratory;

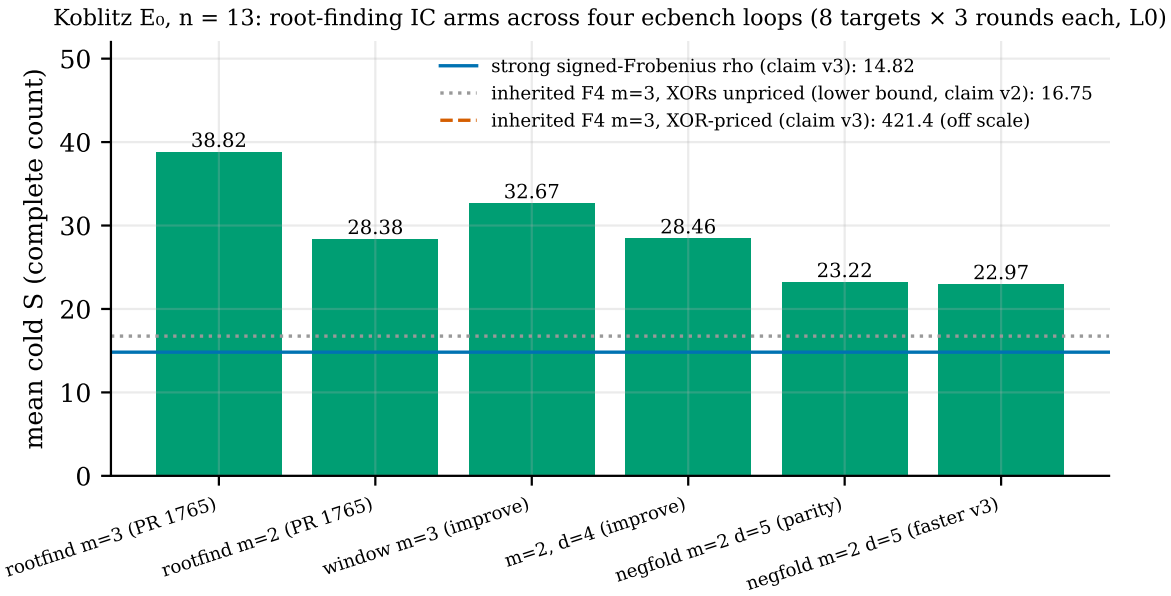


Figure 3: Mean cold S of root-finding IC arms at Koblitz E_0, $n = 13$, $d = 5$ unless stated, across the engine, improve, parity and faster loops. All bars are complete counts. Sources: `research/ecbench_rootfind_engine_20261012/`, `research/ecbench_rootfind_improve_20261013/`, `research/ecbench_rootfind_parity_20261013/`, `research/ecbench_rootfind_faster_20261014/`. Each loop had its own seeds and rho arm (rho 14.12–14.82).

two K1500 runs hit the 900 s timeout and are retained. *Erratum 2026-10-20 (validation sweep B, verdict 110.1: contradicted).* The 18.528 ratio is against rho with an empty table on one target on a shared host. It is superseded and is not an improvement: precomputed-table rho wins every matched $n = 83$ pair. The within-IC K1798 versus K1500 comparison stands.

Reading. Root finding is the largest constant-factor change of the programme at the oracle stage, and the one with a proof behind it. It does not change the asymptotic count of index calculus: its cost is $\Theta(|\mathcal{F}|^{m-1})$, every measurement is at $m \leq 4$, and it leaves relation yield, factor-base size and linear algebra untouched. Its effect on the cold frontier is to move the decomposition stage from dominant to minor, which exposes the stages that now are.

7 Gröbner-basis engines

Before root finding, the decomposition oracle was a Gröbner solve, and for $m \geq 3$ over binary fields it still is. The repository grew a family of engines (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`, §1; file names below are under `src/cryptanalysis/`):

Matrix F4 and F5 over $\mathbb{F}_2$
(`koblitz_groebner.rs`, `matrix_f5_f2.rs`): Macaulay matrices of the Weil-restricted Boolean system, reduced by a shared Four-Russians kernel (`gf2_elim.rs`, four 8-bit tables, AVX-512 or AVX2).

Inherited F4
(`inherited_f4.rs`, from 22 September, the default algebraic oracle): a DPLL-style tree over partial assignments in which each child *inherits* its parent's echelon form and re-reduces only what the assignment changed, instead of building a fresh Macaulay matrix per node.

F6-IC
(opt-in): inherited F4 plus a factor-base domain gate (a fully assigned summand whose abscissa is not a base point refutes; partial support pruning refutes earlier), and meet-in-the-middle leaves (with $m - 1$ summands fixed the last is looked up; at $m = 3$ and $|\mathcal{F}| \leq 256$ every compatible second point is added and the third looked up).

$\mathbb{F}_p$ F4
(`f4_fp.rs`, `f4_fp_tower.rs`, `sig_fp_tower.rs`): degree-bounded F4 with sparse-by-reducer then dense reduction, a Faugère–Lachartre block for the Kummer-tower oracle, and a signature variant.

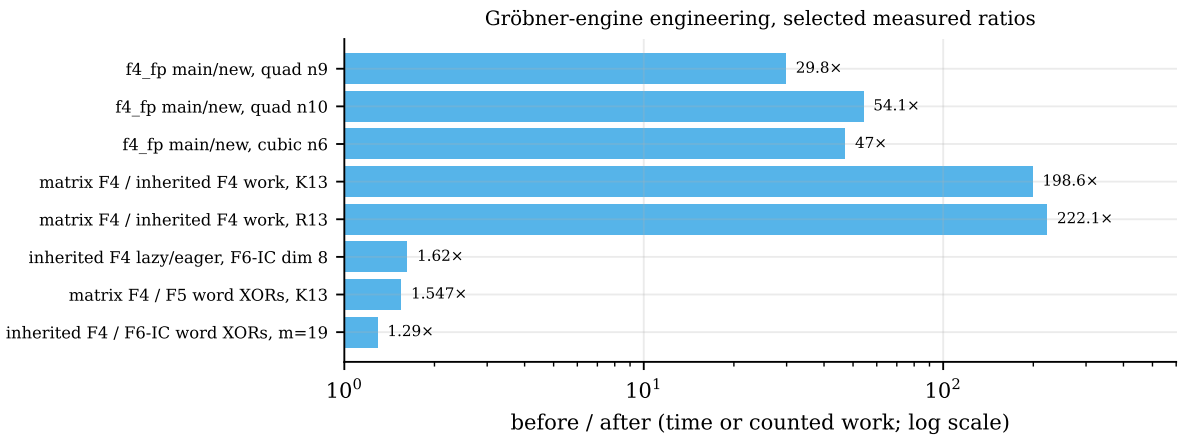


Figure 4: Selected engine ratios (before/after or baseline/candidate): `f4_fp` successive versions (single-threaded wall, L0); matrix versus inherited F4 (counted matrix work); lazy versus eager inherited rewrite (median wall); F4 versus F5 and inherited F4 versus F6-IC (word XORs, stage only). Sources in `figures/data.json`.

Correctness. The audit pinned every engine against every other and against brute force (88+15+9+14+13+12+21+6 passing library tests in the touched modules), and found one soundness bug (§4.5, item 6), two contract defects and several latent holes, all fixed with regression tests.

Engineering results. Figure 4 collects the headline ratios; all are ENGINEERING with identical outputs.

- *Inherited versus matrix F4*: matrix work 198.6× (K13) and 222.1× (random-13) the inherited engine's on identical trees; at $m = 19$ every matrix F4/F5 run timed out at 1,800 s while inherited F4 finished in about 6 s (`research/f4_f5_f6_cross_family_20261011/`).
- *F5 versus F4*: word-XOR ratios 0.6465 / 0.6414 (K13 / random-13), wall 0.79–0.86.
- *F6-IC versus inherited F4*: word XORs about 0.82 at $m = 13$ and 0.77–0.78 at $m = 19$, relation wall 1.7–1.9× faster. But once its batched group additions are charged, F6-IC's S is 64.6–68.8× inherited F4's at $m = 13$ and 1,303–1,353× at $m = 19$: an accounting loss, not a frontier move. At $m = 31$ both engines hit the 3,600 s cap on the first target.
- *Kernel routing and rewrite*: routing every echelon call to the Four-Russians kernel gave 1.75–3.22× on the oracle's own Macaulay matrices; a word-level `pext`/`pdep` rewrite plan made F4 and F6-IC 10–14% faster at $n = 23$; eager rewrite of kept rows beat lazy rows by 1.40–1.62× (`research/inherited_f4_eager_rewrite_20261011/`).
- *Instruction and allocation work*, measured by paired A/B/A′ instruction counts on a frozen eleven-system suite: fused construction and readback, −5.28% (F4) and −11.02% (F5) instructions; native hashing, −3.715% F5 [3.714, 3.726] and −45.1% for the standalone criterion; bounded direct indexing in the F5 packer, −3.947%; row allocation, 42–50% fewer requested bytes. Elapsed time was not claimed for these.
- *Quotient-curve model*: solving the $m = 2$ system directly on the quotient curve instead of a symmetrised control cut F4 word operations by 1.340–1.642× with identical root digests in five interleaved cells (`research/lit2026_w8_summation_models_20261011/`).
- $\mathbb{F}_p$ *F4 to msolve parity* (`research/f4_fp_msolve_parity_20261011/`): Gebauer–Möller criteria, packed monomials, interned symbolic preprocessing and a parallel symbolic pass made `f4_fp` 29.8–54.1× faster than its predecessor on square generic systems, with msolve/ours at 0.96–1.22 single-threaded. A day earlier msolve had been 12–24× faster (`research/gb_cas_comparison_20261011/`); every reduced basis agrees with Singular or msolve. On Boolean decomposition systems ($n = 17$, 8 targets) our oracle decides in 7–24 ms per target, while msolve and PolyBoRi spend 40–52 s computing a full basis and Singular exceeds 120 s—a difference of task (decide versus compute the basis), not of engine quality.

What did not work. Degree-2 early-abort probes after a 2025–26 proposal: word-XOR ratios 1.00083–1.00378 (no change) and 0.99714–0.99928 (worse). Gate-first geometry: 1.57–1.78× fewer word XORs but lane-batched F4 roots cost 2.6–4.0× a sequential control. Convolution splitting of the summation polynomial: 1.21–2.07× *more* word operations despite lower degree; Frobenius auxiliary variables: 57–

701× more (`research/lit2026_w\{4,5,8\}_*_20261011/`). Symmetry-breaking clauses in SAT did not help (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). A registered law for the $m = 3$ refutation degree resolved all 68 units but ended with the verdict "neither": one variant met its generic degree window on 32/32 draws, the other fell below it on 8, and the heaviest degree-10 Macaulay solves were $2.4 \cdot 10^5 \times 2.0 \cdot 10^5$ matrices at 42–51 minutes each (`research/ic_semiregular_law_20261005/`).

8 Koblitz and binary curves

8.1 Frobenius-invariant and compact-orbit bases

The Koblitz implementation (`src/cryptanalysis/koblitz_index_calculus.rs`) builds GGMP's invariant base from the linearised polynomial of a factor of $x^n - 1$, checks $\pi(\mathcal{F}) = \mathcal{F}$ point by point, rewrites each relation into orbit columns with known λ^k weights, and solves the cofactor-corrected system $ha + hbd \equiv \sum \lambda^{k_i} x_{o_i}$. On small fields the orbit collapse is as predicted: $|\mathcal{F}|/\#\text{orbits}$ of 7.86, 10.89, 12.96 at $n = 9, 11, 13$ for relations and 61.7–168.0 for linear algebra, against rho's 4.24–5.10 (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). Because $\ell = \mathrm{ord}_n(2)$ fixes the invariant base's size, the programme moved to bases sized to the instance: divisor bases, SAT-optimised bases, a rank-guided base at $n = 53$ (220 orbits, a complete implicit S_5 formula of 22,887 clauses), and finally the *compact signed-Frobenius orbit* base, which stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). In the ecbench factor-base suite it is the best measured Koblitz construction: $S = 9.114$ at $n = 17$ (272 points, 8 columns), 9.985× matched signed-Frobenius rho, against full orbit 18.7× and subspace bases 47× (`research/ecbench_frontier_20261013/`).

8.2 The oracle sequence

The binary oracle went through SAT (XOR-native CDCL after Trimoska et al. [24]; decides the $n = 19$, $\ell = 6$ corpus but does not scale), matrix F4 (whose refutation range was the real frontier), pairs-and-solve for S_4 (validated to $n = 21$, $\ell = 7$, still $O(2^{2\ell})$), meet-in-the-middle pair tables, inherited F4 and F6-IC (§7), and root-finding engines (§6). The boundary ledger's current Koblitz decomposition record is an $n = 31$, $m = 2$ F4 cell with an exact block-6 M4RI acceleration; at $n = 53$ an exact Frobenius-orbit S_5 compression removes the coordinate ROM but every unrestricted solver arm remains UNKNOWN.

8.3 The one-target online ladder

The ledger's primary `vs_rho` question is: with reusable IC preparation already done, how long does one unseen public target take, against rho with the automorphism discount on the same point? The ladder (Figure 5):

- $n = 53$: guided shared-log precompute with an explicit pair table (9.34 GB peak RSS); earlier, a 1,024-target shared-log batch beat batched rho in all 15 paired blocks (secondary, multi-target).
- $n = 61$: compact orbit, 600 columns, 73,200 points, 2.73 GB; fresh reproduction 124.85× (exploratory).
- $n = 71$: first rung past 64-bit packing (u128 words), 85,200 points, 2,864.1× median of three repeats of one point (exploratory).
- $n = 73$: 87,600 points, 1,189.7×; the guided-rank precompute took 5.1 h at 57–70 million probes per relation (exploratory).
- $n = 83$, $a = 1$: IC online 5.6–7.9 s against rho 43–71 s, 8.2×; then the dense root index, 18.5× (both exploratory; the 18.5× is superseded, an empty-table rho on one target [2]).
- $n = 61$, **panel (13.9x online point superseded 2026-10-20)** (12 October): $a = 0$, $K = 500$ orbit columns, 61,000 points, 64 fresh public one-target workloads per host, identical candidate manifest on two hosts. Pinned x86_64: median online speed-up 13.94×, 95% [9.81, 19.13], claim-check 64/64; unpinned arm64: 12.87× [7.30, 19.08]. Set-up of about 28 s is outside the window; with it charged, IC costs a median 83.4× rho.[3]

The exploratory ratios are large and irregular because a one-point online window measures the luck of one relation search against one rho walk; the admitted panel's range over 64 workloads is 1.06–648.6×.

[2] Erratum 2026-10-20 (validation sweep B, verdict 110.1: contradicted): precomputed-table rho wins every matched $n = 83$ pair.

[3] Erratum 2026-10-20 (validation sweep B, verdict 108.1: contradicted). The 13.94× is against rho with an empty distinguished-point table. It is superseded and is not an improvement: Bernstein–Lange rho at equal advice is 53–317× faster online (§11.3).

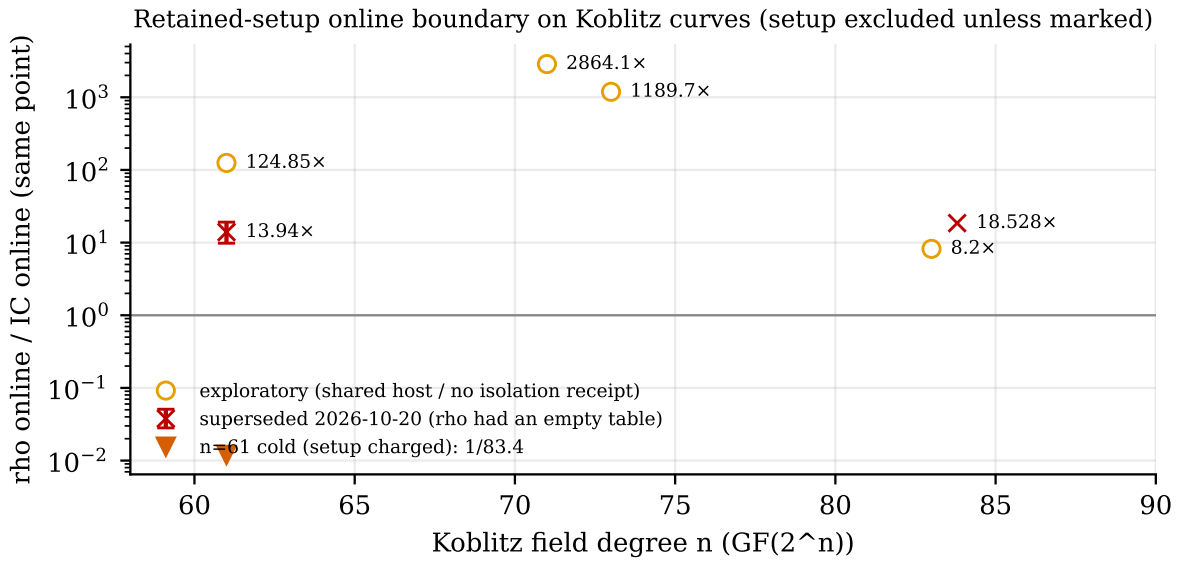


Figure 5: Rho online time over IC online time on the identical public point, reusable IC preparation excluded. Open circles: exploratory (verified answers, no qualifying isolation receipt; three repeats of one point or a single point). Filled square: the admitted 64-workload panel with its 95% bootstrap interval. Triangle: the same $n = 61$ candidate with set-up charged (cold). Sources: `docs/ic/boundary_targets.json`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `research/n83_dense_root_20261013/`.

The admitted point is a constant-factor online result at $n = 61$ (47-bit subgroup) with preparation retained; it is not a cold result, not a scaling law, and not an asymptotic claim. By the frontier's own arithmetic the $n = 83$ preparation would pay for itself after $q^* = 306$ targets (derived what-if: reusable set-up 339.0 s, online 0.0633 versus 1.173 s; assumes constant per-target cost and zero rho set-up).

Retired as an index-calculus win (14 October). Every rung of this ladder compares index calculus with a retained table against rho with none. Given a distinguished-point table of equal or smaller size, Bernstein–Lange preprocessing rho [2] is 53–317× faster online than the compact-orbit arm on the same 64 targets at $n = 41$, 53 and 61 (§11.3). The ladder therefore measures what a table buys, and generic rho buys more per byte.

8.4 ECC2K-130

ECC2K-130 is the $a = 0$, $n = 131$ member of the same family; the compact producer already solves $n = 41, 53, 61, 71, 73$ of it. Extrapolating the fitted probes-per-relation law $C\,r/(n^2K^2)$ with $C = 1$ gives $1.1 \cdot 10^{18}$ core-years at $K = 600$ and $6.8 \cdot 10^{8}$ at $K = 10^{12}$ columns (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`, with the unexplained factor-of-5 gap in C of §4.5). The separate GPU rho estimate for the same curve is about 4.3 RTX PRO 6000 card-years (`paper/ecc2k130/`). Index calculus is not a route to ECC2K-130 on this evidence; the isogeny-class routes are discussed in §10.

9 Prime and extension fields

9.1 Known-answer ladders over $\mathbb{F}_p$

Over prime fields the ledger tracks staged known-answer ladders (`docs/ic/boundary_targets.json`, regime `prime`). On $j = 0$ curves the staged driver recovers a 20-bit logarithm ($p = 1{,}048{,}609$) in 56.8–64.9 s of IC wall against 150–153 ms of rho on the identical public point, with split timers and three seeds. On the generic $a = -3$ shape (P-256 and CryptoPro-B class), a Semaev-S_4 three-decomposition family with Cantor–Zassenhaus root finding yields one relation per trial at $p = 65{,}519$ with a 120-point base; at fixed base size, median trials per relation grow from 2 to 28 to 379–506 at 16, 20 and 24 bits, about 16× per four bits. The prime `vs_rho` row is *not achieved* at any measured size.

9.2 PDP oracle programme

The point-decomposition (PDP) speed-up programme (`research/pdp_oracle_speedup_20261011/`) measured the oracle in three steps before root finding replaced it: a baseline (PS-0; 338–452 ns per pair across five rungs, A/A spread $\leq 8.8\%$), a gate-count model of the group-law form (PS-1; 7,410–14,608 fused ternary-logic issues against 70,695–190,275 for the derived baseline, 9.5–13.0×), and a scalar group-law kernel (PS-2; 29.8–40.8 ns per pair, per-pair ratio to baseline 0.067–0.123, 400 corpus root sets agreeing). A split/hybrid solving-degree study then projected its closest relation phase at 22.8× strong rho (`research/pdp_split_dreg_20261011/`), with an initial read-out defect preserved in the record.

9.3 Kummer-tower factor bases

Petit, Kosters and Messeng's base $V = \mu_{2^t}$ (the roots of $x^{2^t} - 1$) gives prime-field summation systems extra structure for an algebraic solver. We ran it end to end in ecbench with the root-finding oracles on five curves of 20–28 bits (896 runs, all verified, all replay audits OK; `research/pkm_tower_ecbench_20261013/`). Index calculus costs 34.0, 41.9, 29.4, 96.6 and 140.6× rho-negation in S (95% intervals in Figure 6), growing by 0.26 ± 0.28 bits of S_{IC}/S_ρ per bit of r, consistent with the $p^{1/6}$ the floors predict. The tower base costs the same as a plain base of the same size to within 1% (ratios 0.999–1.012): PKM's structure helps an algebraic solver, not a table.

9.4 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$

Over $\mathbb{F}_{p^k}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, a relation needs a k-decomposition solved by a Macaulay system. We made the `gaudry_quartic` solve generic in the number of unknowns and faster (median whole solve 1.9/4.1/6.4× at $d = 4, 5, 6$, 37.4 → 5.9 s at $d = 6$; the S_5 solve from about 30 minutes to 164–203 s; every count and solution identical). Then we built FGHR's D_4 variant (`research/gaudry_quartic_d4_20261013/`): on a curve with rational 2-torsion T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P + T) = -Y(P)$, so S_5 in Y is invariant under the Weyl group D_4 (order 192). In the invariants $\sigma_k = e_k(Y^2)$, $\pi = \prod Y_i$, the system has Bézout number 512 instead of 4,096, Macaulay degree 14 instead of 29, and a $13{,}412 \times 11{,}628$ matrix instead of $41{,}780 \times 40{,}920$.

At $p = 269$, the cost C_4 of one decomposition falls from $1.21\text{–}1.23 \cdot 10^{12}$ to $2.54\text{–}2.55 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications, 47.5–48.4×; a solve takes 4.2–5.6 s instead of 164–203 s. Decompositions agree with an independent meet-in-the-middle enumeration on 19 of 19 solved targets; one target in 20 is degenerate (quotient dimension 517) and is discarded, adding about 5% to the relation phase. The Y-line base decomposes about 1/8 as often (pooled ratio 0.133 over 8,000 targets; prediction $1/8 \pm 30\%$ held on one of two curves), but each column covers $\{\pm P, \pm(P + T)\}$, halving the unknowns.

Inserted into the repository's crossover model (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md` §11.16), the measured C_4 and rate move the $k = 4$ crossover with rho from $2^{149.4}$–$2^{152.7}$ to $2^{135.1}$–$2^{135.8}$ (cofactor $h = 2$, forced by the 2-torsion), and the asymptotic ratio past it from 0.34–0.63 to 0.12–0.22 of rho. This is the one place in the programme where a crossover moves; it moves in an *extrapolated model* ($n^{1/4}$ relations, $n^{1/2}$ linear algebra and rho) fed by a measured constant, so it is classed as ENGINEERING, not an advance. Extension fields of degree four are not the fields deployed curves use.

9.5 Literature claims that did not survive

The lit2026 programme (`research/lit2026_speedup_program/`) ran each finding of a 2025–26 search as a measured change against an interleaved baseline binary. Workstream W12 implemented the zero-minor guess-and-determine attack (Mahalanobis, arXiv 2607.09814v2): it solved 48/48 planted targets on 16–26-bit curves, but its search phases scale with slope 0.905 $[0.769, 1.021]$ in $\log_2 r$ and its cost over rho grows from 260× to 6,715×—exhaustive in r, as the source's own table says. A second flagged ePrint (Hu, 2025/015) did not specify an implementable algorithm and is recorded as blocked. On the generic side, W11 cut BSGS from 16.75 to 3.71 multiplication equivalents per visited point with paired steps (wall 1.15–1.67×), which moves rho's and BSGS's constants, not index calculus.

10 Isogeny, transfer and structure routes

A recurring hope was that some other representation of the same instance—an isogenous curve, a curve at another level of the isogeny volcano, a Jacobian, a Weil descent—would make index calculus cheaper.

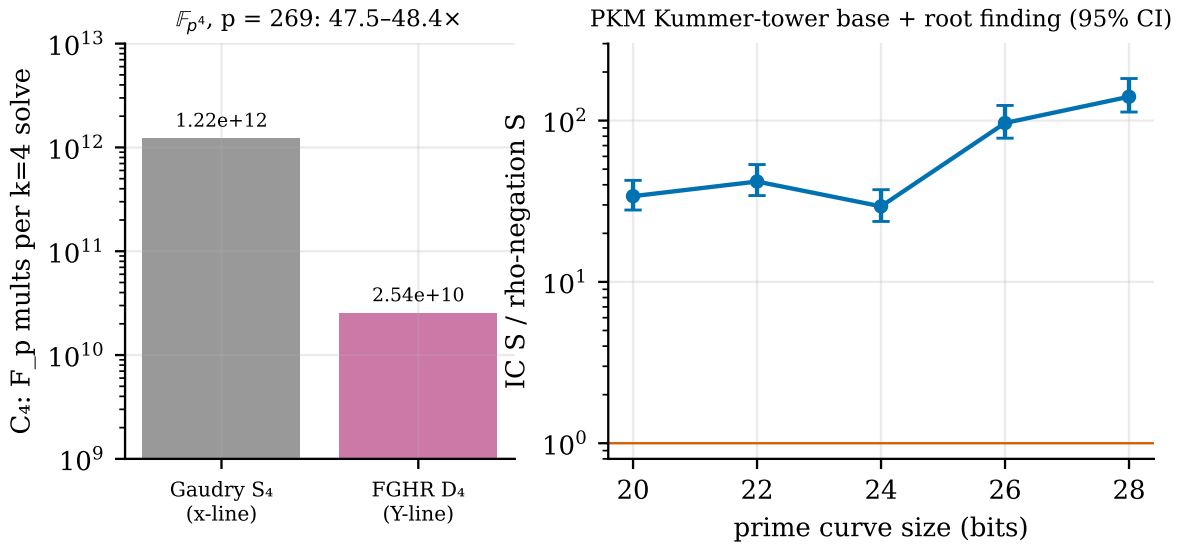


Figure 6: Left: cost C_4 of one $k = 4$ decomposition over $\mathbb{F}_{p^4}$, $p = 269$, Gaudry's S_4 system on the x-line against FGHR's D_4 system on the Y-line (`research/gaudry_quartic_d4_20261013/`). Right: index calculus with the PKM tower base and root finding over rho with negation, eight one-target workloads × three rounds per size, paired 95% cluster-bootstrap intervals (`research/pkm_tower_ecbench_20261013/`).

These routes produced the programme's clearest negatives, and one lemma that explains most of them.

Lemma 1 (transport conservation, `docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §1)**.** *Let $\varphi : E \to E'$ be an $\mathbb{F}_q$-rational isogeny (or any homomorphism of rational points), $\kappa = \ker\varphi \cap E(\mathbb{F}_q)$, $R \in E(\mathbb{F}_q)$, $\mathcal{F}' \subset \varphi(E(\mathbb{F}_q))$ a factor base on the codomain and $\mathcal{F} = \varphi^{-1}(\mathcal{F}') \cap E(\mathbb{F}_q)$. Then the number of ordered m-decompositions of R over $\mathcal{F}$ is $|\kappa|^{m-1}$ times the number of ordered m-decompositions of $\varphi(R)$ over $\mathcal{F}'$.*

Proof. Images of decompositions of R are decompositions of $\varphi(R)$. Conversely, for $\sum Q_i = \varphi(R)$ choose rational preimages P_i of the Q_i; then $\sum P_i - R \in \kappa$, and of the $|\kappa|^m$ choices exactly $|\kappa|^{m-1}$ sum to R (translate P_1 by the kernel element). □

So a homomorphic transfer changes nothing but three curve-specific inputs: the rational kernel symmetry $|\kappa|^{m-1}$ (bounded by the rational torsion; trivial on prime-order curves), the factor-base geometry available on the destination, and the cost of arithmetic there. Sequences of isogenies compose to one homomorphism; a loop is an endomorphism and acts on the subgroup as a known scalar, useful only as a known-log multiplier. Multi-curve factor bases glued by isogenies are single-curve bases in disguise.

Conductor gaps in the ECC2K-130 class. Jao, Miller and Venkatesan [16] prove random self-reducibility among curves with the same endomorphism ring, leaving open whether curves whose conductors differ by a large prime could differ in hardness. The ECC2K-130 class has conductor gap 263; we computed its structure (`docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`) and ran several rounds of experiments. Native counted-S index calculus on two fixed CM pairs gave floor/crater ratios 0.975 (99% $[0.603, 1.451]$) at $\ell = 8209$ and 1.127 $[0.922, 1.342]$ at $\ell = 131101$: no level effect (192 verified records, 96 exact replays; `research/ecbench_conductor_gap_20261012/`). Two-torsion symmetrisation is a quotient isogeny: at a matched base of abscissae it cut F4 row-reduction work 4.1–4.3× ($|H| = 2$) and 17.6–19.6× ($|H| = 4$) over 1,128 systems—but a correction the next day showed that at a matched number of *columns* the saving disappears; it was the cost of a redundant orbit base. A combined Weil descent used 234–529× the F4 operations of the blind system (`research/isogeny_conductor_gap_round3_20261011/`).

Fibre-aware index calculus. Isogeny fibres are torsors whose arithmetic is controlled by one cohomology class (`RESEARCH_ISOGENY_FIBERS_ECDLP.md`); relation generation is isogeny-equivariant at zero lifting cost, and a fibre-closed pair table was the best lever of the conductor-gap rounds (12.8× rho in its best cell). Measured against matched rho across sizes at fixed cofactor, the envelope IC/rho ratio grows

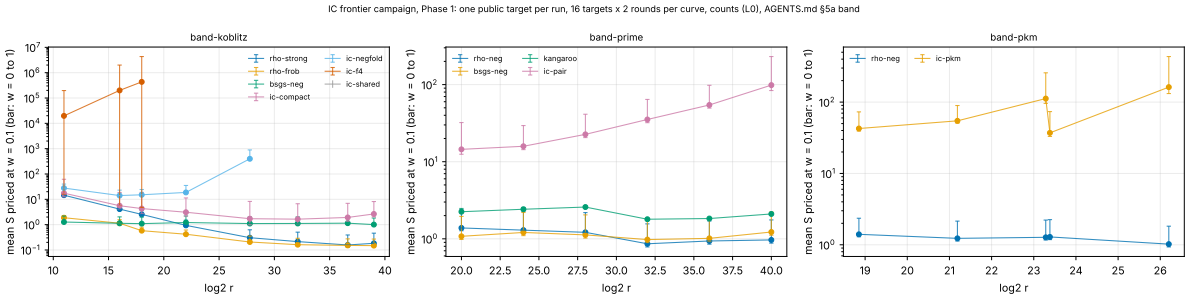


Figure 7: The campaign frontier: mean S per arm and curve at $w = 0.1$ GAE per native operation, the bar spanning $w = 0$ to $w = 1$; one public target per run, 16 targets × 2 rounds per curve, operation counts. Sources: `research/ic_frontier_campaign_20261015/analysis/band-*.json` (`ecbench band`).

from 8.4–8.8 to 35.1–37.0 ($h = 4$) and from 13.4–20.1 to 60.4–63.1 ($h = 72$), as $\#E^{1/6}$: the gap widens, it does not close (816 audited records; `research/fiber_ic_scaling_20261011/`).

Other structure. Symmetrised u-frame bases saved 0.84× at $n = 17$ and 0.76–0.77× at $n = 23$, but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`). GHS and hyperelliptic covers: ECC2K-130 does map to hyperelliptic curves, but the genus that would pay is missing. In odd characteristic the cubic GHS transfer from $E_0(\mathbb{F}_{q^3})$ to a genus-3 Jacobian now runs end to end with double-large-prime index calculus, at about $q^{1.4}$ against rho's $q^{3/2}$, within 1.15× of a measured negation-map rho at $q = 2^{20}$ (§11.3). The isogeny-loop GLV/GLS work produced a measured 1.09–1.15× faster four-dimensional scalar multiplication (`research/isogeny_loop_glv_20261012/`) and a $\sqrt{\text{élu}}$ kernel cost exponent 0.822 $[0.805, 0.838]$; these speed up curve arithmetic, not the discrete logarithm.

Named obstructions. Blocked requirements are kept open as problem records (`problems/README.md`), among them the complete-cost calibration of root-finding IC (`problems/rootfinding-complete-cost-calibration.md`), with exit criteria rather than abandoned.

11 What improved, and where the frontier stands

11.1 The measured frontier: one campaign, every cost priced

The frontier synthesis of 13 October assembled ratios from studies that used different curves, sizes and accounting. To replace it we ran one campaign (`research/ic_frontier_campaign_20261015/`, protocol frozen before any measured run). Every method family ran on the same fixed curve ladders: eight Koblitz curves chosen by subgroup size ($\log_2 r \approx 11$–39, $n \in \{13, 17, 19, 23, 37, 41, 43, 47\}$), six NIST-shaped $a = -3$ prime-order curves of 20–40 bits, and the five 2-adic curves of the Kummer-tower ladder. Each curve had 16 public hash-to-subgroup targets nobody knows the logarithm of, 2 measured rounds and a warm-up, and every answer was checked outside the solver. That is 4,848 executions, 3,232 of them measured: 3,178 verified and 54 exhausted, all of the latter in the shared-rank arm at $n = 41$ and $n = 47$. Every session is audited by re-executing every measured run and requiring identical counts. Index-calculus parameters (compact-orbit columns, negfold subspace dimension, prime base size) were chosen per size on a disjoint set of training targets, with grids widened until the optimum was interior. Every cost is reported across the pricing band of §4.3.

What the campaign shows.

- *No index-calculus arm beats generic rho on any curve, at any weight.* Signed-Frobenius rho is below every index-calculus arm on all eight Koblitz curves and across the whole band; rho with negation is below every prime-field arm on all eleven prime curves (Table 2). The one unresolved pair in the campaign is compact orbit against the *strong* rho reference at $n = 13$, whose scalar-multiplication set-up dominates on an 11-bit subgroup; the cheaper signed-Frobenius walk is below both.
- *Compact orbit is the closest index calculus.* It sits 3.8–15× the best rho with native work free and 4.9–17.8× at $w = 0.1$; the ratio does not shrink with size.

Table 2: Index calculus over the cheapest constant-memory generic arm on the same curve (signed-Frobenius rho on Koblitz curves, rho with negation on prime curves), mean S ratio at three band weights. Every run verified (32/32 per cell). Source: `research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv`.

Curve	$\log_2 r$	Arm	$w = 0$	$w = 0.1$	$w = 1$
Koblitz $n = 13$	11.0	compact orbit	6.9	9.3	27.1
		negfold	14.4	14.8	17.5
Koblitz $n = 17$	16.0	compact orbit	3.8	4.9	11.8
Koblitz $n = 19$	18.0	compact orbit	4.9	7.5	23.6
Koblitz $n = 23$	22.0	compact orbit	5.5	7.4	18.4
Koblitz $n = 37$	27.8	compact orbit	5.2	8.3	23.3
		negfold	1806	1927	2512
Koblitz $n = 43$	32.1	compact orbit	7.4	10.2	23.1
Koblitz $n = 47$	36.6	compact orbit	9.9	12.7	25.3
Koblitz $n = 41$	39.0	compact orbit	15.0	17.8	30.3
prime 20 bit	20	pair table	9.8	10.5	13.5
prime 24 bit	24	pair table	12.1	12.2	12.9
prime 28 bit	28	pair table	18.6	18.6	18.9
prime 32 bit	32	pair table	40.9	40.9	41.1
prime 36 bit	36	pair table	58.1	58.0	57.5
prime 40 bit	40	pair table	95.2	101.8	131.4
PKM 20 bit	18.9	Kummer tower	30.3	30.4	31.0
PKM 28 bit	26.2	Kummer tower	141.3	158.7	238.7

- *Prime-field index calculus falls further behind with size.* The two-summand pair-table arm goes from 10.5× to 101.8× rho (at $w = 0.1$) between 20 and 40 bits, the $r^{1/6}$ widening its $r^{2/3}$ cost against rho's $r^{1/2}$ predicts; the Kummer-tower arm is 28–159× rho.
- *Negfold root finding stops scaling.* It is 12–45× rho up to $n = 23$, about 1,900× at $n = 37$, and does not complete within $5 \cdot 10^7$ trials from $n = 43$ on (PROTOCOL A4b).
- *Inherited F4 is not competitive once its work is priced*: S rises from 16 (word XORs free) to about $2 \cdot 10^4$ at $w = 0.1$ at $n = 13$. The band weighs one 64-bit word XOR like a table lookup, which overstates F4 (its pinned calibration is about 0.002 GAE per XOR); even with XORs free it is above rho.
- *BSGS has the lowest time on several curves* because it spends $\sqrt{r}$ memory; it is reported on the time–memory Pareto set, not ranked against constant-memory methods.

Cost exponents. Fitting total cost $\propto r^{\alpha}$ per arm over each size tier with `ecbench bound fit -weight` (two-stage bootstrap, 95% intervals; a tier needs at least four sizes for an exponent):

- prime pair table $\alpha = 0.620$ $[0.557, 0.663]$ at $w = 0$ against rho with negation 0.435 $[0.331, 0.501]$; separated at $w = 0$ and $w = 0.1$, overlapping at $w = 1$ (0.594 $[0.482, 0.665]$ against 0.444 $[0.334, 0.517]$). Index calculus grows faster than rho (the predicted 2/3 against 1/2), but under the band rule the difference is not established at every weight;
- Koblitz compact orbit, $n = 37$–47: 0.589 $[0.518, 0.743]$ against signed-Frobenius rho 0.436 $[0.358, 0.508]$ at $w = 0$, overlapping at $w = 0.1$ and $w = 1$; on $n = 13$–23 the two overlap at every weight (0.25 against 0.29 at $w = 0$, both with wide intervals because the automorphism count $2n$ changes with the size);
- Kummer tower 0.717 $[0.358, 0.940]$ against 0.438 $[0.332, 0.599]$, overlapping.

No index-calculus arm has a fitted exponent below rho's at any weight, so no crossover with rho is supported by these data, extrapolated or otherwise. The sixty-three fits are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

Limits. Operation counts only (isolation L0); wall time and a second-host replay certificate need the planned RunPod phase. Parameters were selected on four training targets per size. The shared-rank arm leaves three categories of work uncounted and is reported as a lower bound only. Fitted cost exponents per arm and tier are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

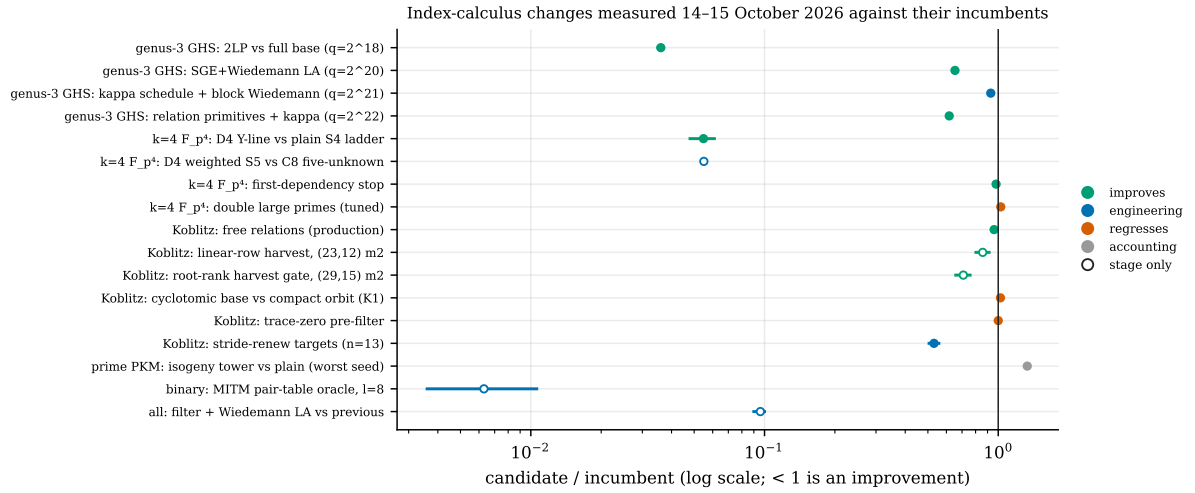


Figure 8: Candidate over incumbent for the changes of 14–15 October, with 95% intervals where the study reports one (the binary oracle's range is the spread over three runs). Filled: whole solve; open: one stage. Values: `papers/ecdlp-index-calculus-program/figures/ladder.tsv`; sources in Table 4.

11.2 The engineering record

Table 3 collects the measured improvements of the programme on the boundary each was measured on. Ratios are baseline/candidate; "stage" means one oracle call or one solver run; "complete" means a verified discrete logarithm with the stated phases.
Derived entries: $53.0 = 1/0.018861$; $17.5 = 1/0.057047$; $1.21 = 1/0.828$ with interval from $[0.783, 0.863]$; $18.3 = 1/0.0545$ with interval from $[0.047, 0.064]$; $1.77 = 300{,}021.9/169{,}069.8$; $1.20 = 1/0.830$ with interval from $[0.697, 0.962]$; $1.87 = 18.714/9.985$ (one session; the compact arm's S is a lower bound, so the ratio is an upper bound); 1.69 and 25.2 compare means from different sessions with different seeds and are descriptive; F5 and F6-IC ratios are reciprocals of the recorded candidate/baseline word-XOR ratios.

11.3 The improvement ladder, 14–15 October

The repository's owner fixed the criterion for success on 15 October: a verified, measured improvement over the current index-calculus baseline is a success; each one becomes the new baseline; the gap to the matched rho is reported as the distance remaining, not as a pass/fail gate; and relabelling or accounting changes do not count. In the two days before that, two improvement rounds and a next-steps campaign landed about twenty measured changes, each paired against its own incumbent under a protocol frozen before data. Figure 8 and Table 4 list them, with the distance to rho that remains after each one. None is in the frontier campaign of §11.1, which predates them, and none was priced across the band; the boundaries are the ones the studies used, which differ from row to row.

Genus three: the one exponent below rho's. The cubic GHS transfer [13] maps the DLP in $E_0(\mathbb{F}_{q^3})$ to the Jacobian of a genus-3 hyperelliptic curve over $\mathbb{F}_q$, where double-large-prime index calculus [14] costs $\tilde{O}(q^{2-2/g}) = \tilde{O}(q^{4/3})$ against rho's $q^{3/2}$. That is known; what is new is a priced, end-to-end, verified run of it on a family the repository can build ($4 \mid \#E_0(\mathbb{F}_{q^3})$, split arm, $q = 2^8$–2^{22}, every logarithm checked against the planted scalar). Adding double large primes changed the exponent from $q^{1.910}$ to $q^{1.385}$ $[1.379, 1.391]$, and keeping the large primes as columns, merging them with structured Gaussian elimination and finishing with Wiedemann moved it to $q^{1.380}$ and the constant to $0.653\times$. This is the only index-calculus family in the programme whose fitted exponent lies below rho's, and the only one where measured IC beats a rho arm: the item-7 rho, which walks at $1.75\sqrt{\ell}$ steps, at $q = 2^{15}$ (0.69 $[0.55, 0.83]$). Two further rounds kept going. A frozen schedule for the factor-base fraction $\kappa(q)$ plus block Wiedemann brought the whole solve to $0.605\times$ the first double-large-prime arm at $q = 2^{21}$. Cheaper relation primitives (one-inversion Cantor addition, a Stickelberger pre-filter, reused exponentiations) then cut the relation phase to $0.47\times$ without changing a single relation, and with κ re-tuned the whole solve to 0.617 $[0.616, 0.619]$ at $q = 2^{22}$, in a unit that also charges inversions. The distance to

Table 3: Measured improvements, July–October 2026. Class: E = ENGINEERING, A = ACCOUNTING. Boundary: St = stage, Cx = complete solve outside ecbench, Cold = ecbench cold one-target S, On = online window after retained set-up. Sources: Appendix A.

Change	Bnd.	Cl.	Ratio	Scope
Decomposition oracle: root finding				
S_3 roots, binary pair table vs matrix F4	St	E	383–1,535	$\mathbb{F}_{2^9}$, $\mathbb{F}_{2^{13}}$, $m = 2$
S_3 roots, prime vs `groebner_f4`	St	E	3.5–4.7·10^4	16–24-bit, $\lvert\mathcal{F}\rvert = 6$
S_3 roots vs Buchberger, NIST curves	St	E	3,021–5,802	P-192. . . P-521
`rf-mitm` vs PS-0 oracle	St	E	3,007–115,160	binary $\ell = 8..14$
Root-finding DLP vs msolve DLP	Cx	E	50–105	16–24-bit prime
Pair-table DLP vs F4 DLP	Cx	E	122	binary $n = 23$ (F4 wins $n \leq 13$)
Pair table vs subtraction	Cold	E	53.0	20-bit prime (bounded)
Frobenius pair table vs subtraction	Cold	E	17.5	Koblitz $n = 17$, priced
Root-finding K13 arm, first to best	Cold	E	1.69	38.82 → 22.97, across sessions
Windowed $m = 3$ scan	Cold	E	1.21	$[1.16, 1.28]$
Negfold root finding vs XOR-priced F4	Cold	E	18.3	$[15.6, 21.3]$, K13
Dense root index K1798 vs K1500 (probes)	On	E	1.77	$n = 83$, one point
Pair-table size 64 vs 32 abscissae	Cold	E	1.20	$[1.04, 1.43]$
Gröbner engines				
Inherited vs matrix F4 (matrix work)	St	E	198.6–222.1	K13, R13
F5 vs matrix F4 (word XORs)	St	E	1.55–1.56	K13, R13
F6-IC vs inherited F4 (word XORs)	St	E	1.22–1.30	$m = 13, 19$
`f4_fp` new vs previous	St	E	29.8–54.1	square systems, = msolve
Four-Russians routing	St	E	1.75–3.22	oracle matrices
Eager vs lazy inherited rewrite	St	E	1.40–1.62	dim 8–9
Quotient-curve $m = 2$ model	St	E	1.34–1.64	five cells
Factor bases and symmetry				
Compact orbit vs full orbit	Cold	E	≤ 1.87	Koblitz $n = 17$; compact is LB
FGHR D_4 vs Gaudry S_4, C_4	St	E	47.5–48.4	$\mathbb{F}_{269^4}$
`gaudry_quartic` solve	St	E	1.9–6.4	$d = 4..6$
One-target online				
Compact-orbit IC vs strong rho	On	E	13.94	$n = 61$; rho without a table; retired (§11.3)
Accounting (costs that went up when priced)				
F4 word XORs priced	Cold	A	1/25.2	16.75 → 421.4, K13
F6-IC batched geometry priced	Cold	A	1/64.6–1/1,353	$m = 13, 19$

Table 4: Changes measured 14–15 October 2026 against their own incumbents. Ratio: candidate/incumbent, < 1 better. Class, against the index-calculus incumbent (not the paper's ADVANCE, which is against rho): Ie = improves the exponent, Ic = improves the constant, E = ENGINEERING on one stage, R = regresses, T = trades, A = ACCOUNTING. "To rho": the distance left to the matched rho on the study's own boundary; "d" marks a derived or extrapolated value. Sources under `research/`: `ic_next10_20261014/`, `ic_improve_20261014/`, `ic_improve_20261015/`, `ic_improve3_20261015/` and the named study directories.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer, $E_0(\mathbb{F}_{q^3})$, cost in $\mathbb{F}_q$ multiplications			
Double large primes vs full base (04)	Ie	0.036 $[0.035, 0.036]$ at 2^{18}; $q^{1.910} \to q^{1.385}$	2.58× negation rho model at $q = 2^{20}$
Large primes as columns, SGE, Wiedemann (01)	Ie	0.653 $[0.647, 0.660]$ at 2^{20}; $q^{1.398} \to q^{1.380}$	—
$\kappa(q)$ schedule + block Wiedemann (06)	T	0.929 at 2^{21}	1.15× measured negation rho at 2^{20} (ref.)
Cheaper relation primitives + re-tuned κ (r3-01)	Ic	0.617 $[0.616, 0.619]$ at 2^{22}	no matched rho in this unit
$k = 4$ over $\mathbb{F}_{p^4}$, operation counts			
D_4 Y-line vs plain S_4, $p = 101$–1033	Ic	0.0548 $[0.048, 0.061]$	handover 2^{126}–2^{137} (d)
D_4-weighted S_5, four unknowns, vs five	E	1/18.5–1/17.8	handover $2^{135.6} \to 2^{118.9}$ (d)
First-dependency stop (`d4-ed`)	Ic	0.979 $[0.973, 0.986]$	no $k = 4$ run below rho
Double large primes, tuned	R	1.025 $[1.016, 1.037]$ (LA 0.827)	—
Koblitz, compact orbit and Gröbner oracle			
Free relations in production, $n = 41, 53, 61$	Ic	0.960 $[0.946, 0.976]$	3.8–15× rho-frob ($w = 0$, §11.1)
Linear-row harvest, $(23, 12)$, $m = 2$	Ic	0.858 $[0.803, 0.916]$	oracle stage; default on
Root-rank harvest gate, $(29, 15)$, $m = 2$	Ic	1/1.41 $[1/1.52, 1/1.32]$; end to end 1/1.20	oracle stage; default on
Gated fall-row augmentation	Ic	1/11.9 on barrel $n = 13$; 1 in production	production no-op
Equivariant Macaulay oracle	T	4.0 ($n = 5$) … 1/15.5 ($n = 13$)	not applicable to production
`stride-renew` targets, $n = 13$	E	0.531 $[0.506, 0.557]$	4.7× rho-strong at $n = 19$
Cyclotomic γ-weight base, K1	R	1.022 $[1.021, 1.023]$	yield-neutral
Trace-zero pre-filter	R	1.00003	vacuous on odd order
Prime fields and shared stages			
PKM isogeny tower, search charged	A	1.002–1.331	17–137× rho
Binary MITM pair-table oracle, $\ell = 8$	E	1/277–1/94 per target	no exponent claim
Filter + Wiedemann LA, $D = 2^{18}$	E	1/10.40; 1/1.93 vs CADO-NFS	LA is 0.1–1.9% of runs

rho shrank from 2.58× the negation-map rho model at $q = 2^{20}$ (first double-large-prime arm) to 1.15× a measured negation-map rho there after the second round ($2.31 \cdot 10^{10}$ against $2.01 \cdot 10^{10}$ $\mathbb{F}_q$ multiplications; a reference on overlapping but not identical curves). The third round's 0.62× has no matched rho in its unit yet, so whether it crosses is not established. The local exponents sit at 1.38–1.41, still above 4/3, because the linear algebra grows faster than $q^{4/3}$ and the tuned κ falls with q. The family is narrow: the transfer needs the right 2-power torsion, and $\mathbb{F}_{q^3}$ curves are not deployed. *Next improvement*: a negation-map rho measured in the third round's unit on the same instances, then the run past $q = 2^{22}$ that would show a crossover or rule it out.

The $k = 4$ chain. The D_4 Macaulay system of §9.4 entered a complete ladder ($p = 101$–1033, 21 curves): 18.2× cheaper than plain S_4 at the same exponent, with the handover with negation rho extrapolated to 2^{126}–2^{137}. Writing the D_4-symmetrised S_5 in four weighted unknowns cut the per-solve cost another 17.8–18.5×, to $1.42 \cdot 10^9$ $\mathbb{F}_p$ multiplications, 854× below the plain form; in the 2-torsion frame that moves the derived handover from $2^{157.9}$ to $2^{118.9}$. Double large primes cut the linear algebra (0.827×) but regress the whole solve; stopping at the first dependency gains 2.1%. No $k = 4$ run below rho exists; every handover is an extrapolation of fitted exponents through a measured constant. *Next improvement*: put the four-unknown solve into the complete ladder, then keep double large primes' linear-algebra gain without its relation-phase cost.

Koblitz. Free relations—root-table self-collisions used as homogeneous rows—are now the production default (0.960× counted, 0.941× wall); the linear rows the Gröbner oracle used to discard are harvested (0.86–0.93× where the gate fires, and a root-rank gate then made it fire on more cells: 1/1.41 at $(29, 15)$, 1/1.20 end to end); the gated fall-row and equivariant passes pay only on overdetermined systems the production pipeline does not build. The cyclotomic base and the trace-zero filter were refuted. A census closes the free-relation lever at the rungs run: at $n = 61$, $K = 600$ the self-collisions span 49 independent relations of the 599 needed, and full rank would need $K \approx 2{,}050$. The distance to signed-Frobenius rho is still 3.8–15× cold at $w = 0$, and the fitted Koblitz ratio grows as about $r^{1/6}$ (slope 0.165 $[0.117, 0.213]$ per $\ln r$ on eight public curves, every native priced). *Next improvement*: relation yield per probe, which none of these changes touched.

The online result is retired. An equal-advice audit gave Bernstein–Lange preprocessing rho [2] a distinguished-point table no larger than the index-calculus pair table and ran both on the same 64 unseen targets at $n = 41$, 53 and 61. Rho with the table is 53–317× faster online (at $n = 41$, with exactly equal advice, 317 $[137, 739]$), and even at 1/1,024 of the bytes it stays 7.7–32× faster. The 13.9× online win of §8 and the later 7.3× and 12.3× hosted single-target points at $n = 71$ and $n = 83$ compared index calculus *with* a table against rho *without* one. They are measurements of the preprocessing model, not index-calculus improvements, and the online frontier belongs to preprocessing rho. The 18.5× dense-root point at $n = 83$ is retired on the same ground (erratum 2026-10-20, validation sweep B, verdict 110.1: empty-table rho on one target). The 7.3× and 12.3× points were also corrected: the ratios of mean online times are 4.21× and 2.80×, and precomputed-table rho wins every matched pair (verdict 104.1).

The reference moved too. Two rho studies shortened the distances above. On prime curves, choosing the better of $P + R$ and $P - R$ at each step takes 1.129× fewer steps under distinguished points (1.081–1.184); each step costs more, so the predicted net gain is 1.03–1.12× with 8–64 walks per inversion and a net loss (0.95) at 1,024. On ECC2K-130, signed σ steps cut GPU time per solve to 0.961–0.970 (derived; not rolled out). Neither is yet in a campaign arm. A comparison against a rho that has not adopted them would overstate index calculus's position by that much.

11.4 Three frontiers

Cold, one target, set-up included. Unchanged: generic rho. Every measured index-calculus arm is above its matched rho (Figure 9); the closest complete count is root-finding negfold at 1.55× strong signed-Frobenius rho at $n = 13$; the closest lower-bound arms are 1.08× (compact orbit with a counted Frobenius table) and 6.97× (matrix F4/F5 against rho-negation at K13). Prime-field arms are 12–239× rho and diverge with size.

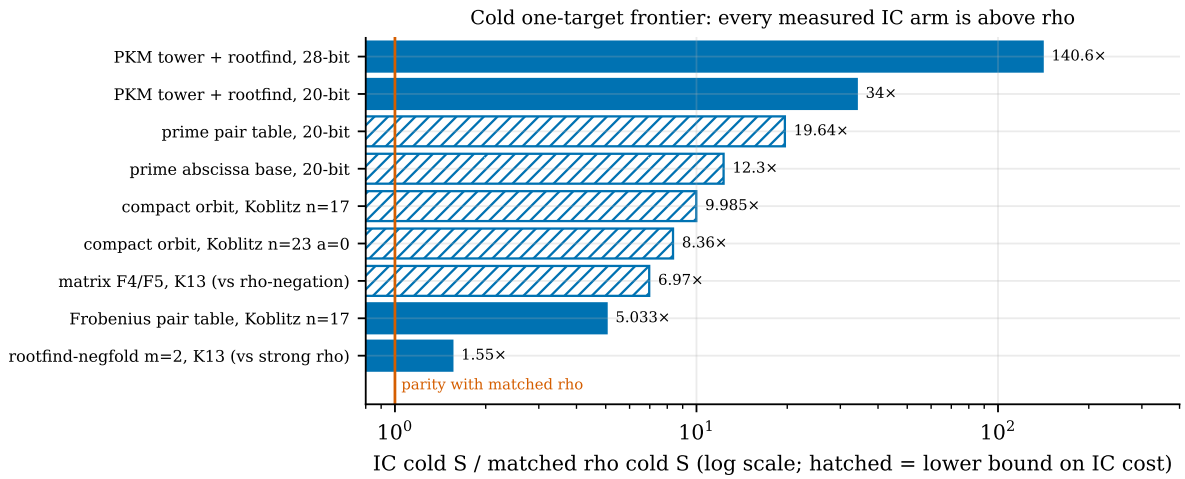


Figure 9: Cold one-target S of index-calculus candidates over their matched rho arm (each study's own estimator; curves differ, so bars are not ranked against each other). Hatched: lower bound on IC cost (unpriced native or solver work). Sources: `research/ecbench_frontier_20261013/`, `research/ecbench_rootfinding_assessment_20261012/`, `research/pkm_tower_ecbench_20261013/`, `research/ecbench_rootfind_faster_20261014/`.

Online, after retained set-up. Generic again: Bernstein–Lange preprocessing rho. The compact-orbit point at Koblitz $n = 61$ (admitted when measured, now superseded, erratum 2026-10-20, validation sweep B, verdict 108.1), 13.9× [9.8, 19.1] over strong rho without a table (cold median 83.4× rho), is 53–317× slower online than preprocessing rho given equal or less advice (§11.3).

Stage. Moved substantially: the decomposition oracle is two to five orders of magnitude cheaper than at the start of the programme on the workloads where root finding applies, the $\mathbb{F}_p$ F4 is at msolve parity, the Boolean F4 family is one to two orders of magnitude cheaper in counted matrix work, and the $k = 4$ extension-field solve is 48× cheaper. None of this, by itself, moves the cold frontier: the stages it leaves untouched—relation yield, factor-base size, linear algebra, table set-up—now dominate.

11.5 What would move the cold frontier

From the frontier synthesis and the problem records: (i) register the torsion-symmetrised bases and the $n = 83$ dense-root engine in ecbench under immutable method identities and emit cold and online records; (ii) resolve the inherited-F4 cross-host counter drift before using that counter in a bound; (iii) run a four-size native challenge against the standing rho bound—two sizes cannot establish an exponent; (iv) obtain an isolated cold session at $n = 83$ with every set-up phase and retained byte charged; (v) price F6-IC's batched geometry with a pinned conversion. A result in the cold domain that beats rho at four sizes would be the first ADVANCE of the programme.

12 Open problems and proposed experiments

1. The solving degree of Weil-descended summation systems. Every subexponential claim for prime-degree binary curves rests on a bound for this degree (§3); none is proved and the experimental evidence is contested. *Experiment:* measure first-fall, last-fall and solving degree of S_{m+1} restricted to ℓ-dimensional subspaces over a grid of (n, ℓ, m) with $m \geq 3$, at least four values of n per (ℓ, m), against the semi-regular prediction, with multi-graded regularity (one block per summand) as a second model. *Settles:* whether the degree grows with n at fixed (ℓ, m); a growth consistent with zero would revive the Petit–Quisquater and Semaev heuristics, a positive slope would close them for this model.

2. Cold index calculus against rho with every operation priced. No measured arm beats matched rho on the cold one-target boundary once native work is priced on both sides (§11). *Experiment:* the frontier campaign of §11 extended to the largest sizes each arm can reach, with exponents fitted per arm and the crossover reported only where the fitted exponents separate across the whole pricing band. *Settles:* whether any implemented arm has an exponent below 1/2 in r on these families.

3. Relation yield is now the dominant stage. Root finding made the decomposition oracle cheap; what remains is the number of trials per relation, governed by the factor-base density. *Experiment:* for each factor-base family, the measured trials-per-relation law against r and base size, compared with the counting prediction $m!\,r/N^{m-1}$; the residual (e.g. the unexplained factor 5 in the ECC2K-130 extrapolation, §8) is the target.

4. Root finding beyond two summands on prime curves. The two-summand oracle is prior art in mechanism (McGuire–Mueller) and costs $\Theta(N)$ per trial; three summands cost a pair table of N^2 entries. *Question:* is there a sub-N^2 structure for S_4 membership over a prime-field base—for example from the D_m-invariant coordinates of 2-torsion curves—and does it change the $r^{2/3}$ cost of two-summand prime index calculus? A negative answer with a counting argument would be as useful as a positive one.

5. The preprocessing regime. Index calculus with a retained table beats rho without one, and loses by 53–317× to Bernstein–Lange rho with an equal or smaller table (§11.3). *Experiment:* measure the advice–time curve (S, T) of the compact-orbit arm across sizes against preprocessing rho's $ST^2 \approx N$; an index-calculus online result counts only if it lies below that curve at equal S.

6. Symmetry at $k = 4$, end to end. The D_4 reduction and its four-unknown weighted form move a *modelled* handover over $\mathbb{F}_{p^4}$ from about 2^{150} to 2^{119}–2^{137} depending on frame (§11.3). *Experiment:* a complete $k = 4$ index calculus over $\mathbb{F}_{p^4}$ at several p, cold, priced, against rho with the available automorphisms, to replace the model by a measured slope.

7. Pricing and calibration infrastructure. Rankings now require every arm's native work to be counted (AGENTS.md §5a). *Work:* count the three uncounted categories of the shared-rank arm; calibrate every ladder size on one fixed host; resolve the inherited-F4 cross-host counter drift; register the $n = 83$ dense root index in the harness.

8. Genus three after cubic GHS, to the crossover. The one family whose fitted exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$ was 1.15× a measured negation-map rho at $q = 2^{20}$ before a further 0.62× step (§11.3); its local exponents stay near 1.4 because the linear algebra grows faster than $q^{4/3}$. *Experiment:* measure negation-map rho on the same instances in the latest unit (inversions and mod-ℓ multiplications priced), and run both past $q = 2^{22}$, in the harness and across the pricing band. A measured crossover there would be the programme's first ADVANCE against rho, on a narrow and undeployed family.

13 Conclusion

Fourteen weeks of work produced three kinds of result. The first is infrastructure: a single unit and harness in which rho, BSGS, the kangaroo and every index-calculus variant are charged the same way on the same one-target workloads, with sealed records, exact replays and admitted intervals; and a ledger that refuses a claim with a missing field. The second is a set of constant-factor improvements, the largest of which—the closed-form roots of S_3—began as one sentence in a manuscript we were asked to check, and ended as a proved identity, an oracle family covering prime and binary fields, two to four summands, CPU and GPU, plain, tower and dense-index bases, and the default decomposition strategy of the codebase. The third is a map of what does not work, which is longer than the list of what does: SAT that does not scale, symmetry that pays at $n = 17$ and costs at $n = 31$, transfers that provably cannot change decomposition counts, fibre-aware bases whose gap to rho grows as $\#E^{1/6}$, literature attacks that are exhaustive search.

The honest summary is the campaign of §11.1. On the question the survey asks—does index calculus solve one unseen elliptic-curve target, cold, for less than rho?—the answer on every curve we measured, at every weight of the pricing band, is no. The closest arm is a compact Frobenius-orbit Koblitz index calculus at 4–18× signed-Frobenius rho, prime-field index calculus falls behind as $r^{1/6}$, and no fitted exponent of an index-calculus arm lies below rho's. The online 13.9× at $n = 61$ compared a table against no table and loses 53–317× to preprocessing rho with equal advice; it is retired.

Measured against its own baseline, which is the criterion the programme now works to, index calculus keeps improving: about twenty changes landed on 14–15 October, among them a genus-3 index calculus after cubic GHS whose exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$, an 18× cheaper $k = 4$ ladder and a further 18× cheaper $k = 4$ solve, and production free relations on Koblitz curves (0.960×). The genus-3

family is the nearest to rho: 1.15× a measured negation-map rho at $q = 2^{20}$ after the second round, with a further 0.62× step not yet matched against rho. The stages that now dominate are relation yield, factor-base size and linear algebra, and the open problems of §12 say what would move them.

A Artefact index

Paths are relative to the root of the `crypto` repository. Each study directory holds its protocol, raw records, run reports with graphs and a PDF; ecbench studies also hold sealed sessions and audit receipts.

Measurement
`docs/ecbench/README.md`; `src/cryptanalysis/ecbench/`; `docs/ic/BOUNDARY_TARGETS.md`; `docs/ic/boundary_targets.json`; `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`; `docs/bounds/frontier.json`; `research/ecbench_frontier_20261013/`.

Root finding
`RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`; `papers/rootfinding-decomposition/`; `research/ecbench_rootfinding_assessment_20261012/`; `research/semaev_root_vs_groebner_20261012/`; `research/semaev_rootfind_nist_20261012/`; `research/semaev_rootfind_e2e_20261012/`; `research/pdp_root_finder_20261012/`; `research/pdp_root_finder_ecbench_20261013/`; `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`; `research/ecbench_rootfind_engine_20261012/`; `research/ecbench_rootfind_improve_20261013/`; `research/ecbench_rootfind_parity_20261013/`; `research/ecbench_rootfind_faster_20261014/`; `research/ecbench_rootfind_vs_rho_20261014/`; `research/ecbench_rootfinding_tuning_20261013/`; `research/ecbench_rootfinding_arity_20261013/`; `research/ecbench_rootfinding_batchtable_20261013/`; `research/ecbench_rootfinding_lanes_20261013/`; `research/n83_dense_root_20261013/`.

Gröbner engines
`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`; `research/f4_f5_f6_cross_family_20261011/`; `research/inherited_f4_eager_rewrite_20261011/`; `research/f4_f5_fused_readback_20261012/`; `research/f4_f5_hash_20261012/`; `research/f4_f5_dense_packing_20261012/`; `research/f4_f5_row_build_20261011/`; `research/f4_fp_msolve_parity_20261011/`; `research/gb_cas_comparison_20261011/`; `research/ic_semiregular_law_20261005/`; `research/f6_ic_ecbench_ladder_20261005/`.

Koblitz
`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`; `RESEARCH_ECC2K130_IC_FEASIBILITY.md`; `docs/ic/COMPACT_ORBIT_PRODUCER.md`; `research/ecbench_factor_bases_20261013/`; `research/koblitz_symmetrised_e2e_20260927/`.

Prime and extension fields
`research/pdp_oracle_speedup_20261011/`; `research/pdp_split_dreg_20261011/`; `research/pkm_tower_ecbench_20261013/`; `research/gaudry_quartic_d4_20261013/`; `research/gaudry_quartic_speed_20261013/`; `research/lit2026_speedup_program/`; `research/lit2026_w12_flagged_claims_20261011/`; `research/lit2026_w11_generic_engineering_20261011/`.

Transfer and isogeny
`docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md`; `docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`; `RESEARCH_ISOGENY_FIBERS_ECDLP.md`; `research/ecbench_conductor_gap_20261012/`; `research/isogeny_conductor_gap_round3_20261011/`; `research/fiber_ic_scaling_20261011/`; `research/isogeny_loop_glv_20261012/`.

This paper
`papers/ecdlp-index-calculus-program/figures/data.json` (every plotted value with its source); `figures/make_figures.py`; browsing copies in `benchmarks/20261014-ic-program-paper/`.

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